Rock Mechanics and Rock Engineering
https://doi.org/10.1007/s00603-026-05868-3

ORIGINAL PAPER

Heterogeneous Response and Zonal Micromechanical Deterioration of Deep Rock-forming Minerals Under Laser Irradiation

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Received: 17 March 2026 / Accepted: 28 July 2026
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Abstract

The mineralogical heterogeneity of deep hard rock is a critical factor governing the efficiency of laser-assisted drilling. This study investigates the micro-scale response differences of four constituent minerals, including fluorite, quartz, tourmaline, and plagioclase, by categorizing laser-affected zones into "weak, moderate, and strong" regions along the radial direction from the beam center. Utilizing an integrated characterization framework consisting of infrared thermography, 3D laser scanning, X-ray diffraction, and nano-indentation, we elucidated the thermal response, phase transformation, and spatial attenuation of mechanical properties. The findings reveal: (1) Fluorite, tourmaline, and plagioclase exhibit exceptionally high laser sensitivity, with spot center temperatures rapidly escalating to 925°C, triggering intense vaporization and melting. Conversely, quartz demonstrates superior structural stability, where energy dissipation is predominantly governed by thermal conduction. (2) Damage features are primarily controlled by the Gaussian energy distribution of the laser. Fluorite and plagioclase develop significant elevation gradients, while tourmaline undergoes typical brittle degradation. Quartz exhibits phase transformation only within the localized peak-energy region. (3) High-intensity laser thermal effects induce the enrichment of low-energy stable phases such as quartz. The primary component of fluorite decreases sharply by 54.5%, accompanied by the selective volatilization of impurity phases such as calcite. (4) Nano-indentation confirms that mechanical parameters attenuate with increasing radial distance. The most pronounced weakening occurs in plagioclase and fluorite, with peak reductions exceeding 80%, whereas quartz maintains significant mechanical resilience. Notably, tourmaline exhibits distinct curve dispersion and "pop-out" events during indentation, substantiating the occurrence of severe brittle damage. These insights provide a theoretical foundation for developing adaptive laser pre-weakening strategies based on real-time lithological identification in hard rock engineering.

Highlights

1. Deciphered the laser-rock interaction mechanisms based on the intrinsic thermal-physical response of individual rock-forming minerals.
2. Quantified the nano-scale morphological evolution and mechanical deterioration (Hardness and Elastic modulus) using high-resolution SEM and nano-indentation mapping.
3. Validated the spatial zoning of laser-induced damage from a micromechanical perspective, confirming the radial attenuation of mineral strength.
4. Established a hierarchy of laser sensitivity for deep minerals, providing a theoretical foundation for adaptive laser-assisted drilling.

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Published online: 13 August 2026

Keywords Laser-rock interaction · Rock-forming minerals · Nano-indentation · Micromechanical deterioration · Zonal characteristics

1 Introduction

As global energy exploration advances into deep and ultra-deep formations exceeding 4500 m, deep-seated oil and gas resources have become pivotal for ensuring national energy security, given that their reserves are estimated at 671 billion tons of oil equivalent and account for 34% of the global total. However, the extreme "Three-Highs" environment including high geo-temperature, high in-situ stress, and high fluid pressure characteristic of deep strata plunges conventional mechanical drilling into the dual predicament of functional instability and soaring operational costs (Lu et al. 2025a). In deep-seated dense hard rock formations, the Rate of Penetration (ROP) of traditional drilling technologies often drops below 1 m/h, while the service life of drill bits is significantly curtailed to less than 50 h (Lukawski et al. 2014; Jamali et al. 2019). Furthermore, explosive blasting in deep mining operations frequently triggers severe rock bursts and excessive damage to the surrounding rock mass due to the uncontrolled nature of its energy release (He et al. 2023). Consequently, the development of high-efficiency, high-precision, and low-disturbance non-contact hard rock fracturing technology has emerged as a critical scientific bottleneck that demands an immediate breakthrough in the field of deep earth engineering.

Laser-assisted rock breakage technology, characterized by high energy density, rapid fracturing, and precise controllability, offers an innovative solution for the efficient excavation of deep hard rock (Zhou et al. 2025). Existing studies have demonstrated that laser irradiation can significantly weaken the macro-mechanical strength of rocks (Yang et al. 2023; Xu et al. 2024; Gao et al. 2024). Considering the high energy consumption associated with standalone laser drilling, Hartlieb and Bock (2018) proposed a synergistic approach combining "laser pre-treatment" with mechanical breakage. The essence of this strategy lies in utilizing laser energy to pre-weaken the rock structure, thereby enhancing mechanical breaking efficiency while reducing tool wear and overall operational costs. Unlike the instantaneous energy release of explosive blasting, laser breakage originates from the interaction between high-energy photons and the rock mass. By focusing a laser beam with extreme energy density onto a target area within a millisecond timeframe, efficient rock removal is achieved through fragmentation, melting, and vaporization (Bakhtbidar et al. 2020). Experimental investigations by Chen et al. (2022a, b) indicated that the integration of an elliptical laser beam into drilling operations led to a substantial increase of 61% in the Rate of Penetration (ROP). Furthermore, numerical simulations conducted by Tao et al. (2025) validated the positive contribution of thermal stress-induced crack initiation to the overall effectiveness of assisted rock breaking. Recent work by Sinha and Joshi (2026) reported mode-dependent thermal zones and surface morphologies under laser scanning and burst-mode pulsing.

As a heterogeneous assemblage of various rock-forming minerals, the macro-mechanical behavior of rock is fundamentally governed by the physical properties of its microstructural units (Hemmati et al. 2020; Ma et al. 2021; Göğüş et al. 2024). Under laser irradiation, individual minerals exhibit pronounced non-uniform thermal response characteristics due to significant disparities in optical absorption, thermo-physical parameters, and mechanical strengths (Lasheen et al. 2023; Yang et al. 2024; Ma et al. 2025). Yang et al. (2020) identified quartz content as a pivotal factor in the staged temperature leaps during laser perforating. Specifically, as the quartz fraction increases, the upper temperature limit of the rapid-rise stage elevates significantly, the gradual-rise stage extends with a slower heating rate, and the final equilibrium temperature increases accordingly. This phenomenon represents a typical non-uniform, step-wise thermal response, driven by the differentiated melting mechanisms of constituent minerals. Further investigations indicate that the melting characteristics of single-mineral phases, such as quartz and feldspar, dominate the thermal equilibrium and temperature evolution within the perforation zone. Owing to its higher melting point and refractoriness, quartz-rich rock masses exhibit higher thermal stability limits and equilibrium temperatures. Thus, quartz content and its associated thermo-physical parameters constitute the primary controlling factors for non-uniform heating and phase transformation behavior under laser exposure. Furthermore, the superposition of the deep in-situ stress field and laser-induced thermal stress further triggers complex damage evolution, including dislocation slip and grain boundary separation. Integrating CT three-dimensional reconstruction, Zhou et al. (2026) discovered that the confining pressure level regulates crack nucleation mechanisms, shifting the failure mode from unidirectional tensile cracking to a complex, multi-directional instability within the fracture network.

Despite progress in macro-scale experimentation, traditional destructive testing remains inadequate for accurately characterizing the evolution of micromechanical responses at the mineral scale, primarily due to specimen size effects and the inherent difficulty in isolating high-purity mineral phases (Konrad-Schmolke et al. 2018; Keith et al. 2023). Fundamentally, deep hard rock is a multiphase composite comprising heterogeneous minerals, pores, and organic components; its

macro-mechanical behavior represents an averaged manifestation of these diverse microscopic attributes (Li et al. 2025; Waqar et al. 2025). To unravel the mechanisms of high-energy rock fragmentation, the research paradigm is shifting from macro-scale observations toward in-situ micromechanical characterization. Nano-indentation technology, with its superior spatial resolution, effectively overcomes the sample size constraints associated with conventional testing, enabling the precise probing of individual mineral grains. Bai et al. (2024) utilized this technique to quantitatively characterize the deterioration of mineral hardness and modulus following microwave irradiation. Their work robustly validates the reliability of nano-indentation in deciphering the micromechanical damage evolution of rocks under extreme physical fields, providing a crucial methodological foundation for the present study.

However, a significant scale gap persists in current research. While macro-scale experiments have extensively focused on the influence of laser parameters on failure morphology, direct observations of mechanical deterioration patterns at the mineral scale remain scarce. Consequently, existing micromechanical insights are insufficient to support robust theoretical predictions of rock breakage behavior in complex deep-earth environments (Sharma et al. 2018; Wang et al. 2020; Chen et al. 2022b; Guo et al. 2022; Pan et al. 2024; Lu et al. 2025b; Zhang et al. 2025; Liu et al. 2025). This deficiency in fundamental understanding hinders the transition of laser-assisted drilling technology from laboratory exploration to field-scale engineering applications. Laser-induced rock failure originates from the degradation of mineral microstructures, which subsequently evolves into the initiation and coalescence of macro-fractures. Nevertheless, a comprehensive understanding of the thermal damage characteristics and the evolutionary patterns of mechanical deterioration across diverse minerals in deep rock masses remains elusive. In this context, this study integrates X-ray diffraction (XRD) phase analysis, ultraviolet–visible-near-infrared (UV–Vis-NIR) spectroscopy, scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM–EDS) observations, and nano-indentation testing to systematically investigate the surface thermal damage and nanomechanical behavior of four representative rock-forming minerals subjected to laser irradiation. This work aims to refine the multi-scale theoretical framework of laser-rock interactions, providing a solid theoretical foundation for the development of high-efficiency drilling technologies for deep-seated resources.

2 Materials and Methods

2.1 Experimental Design and Procedure

To unravel the micro-scale damage mechanisms of deep rock-forming minerals under laser irradiation, a sequential experimental framework consisting of "initial physical property calibration -laser irradiation-multi-perspective damage evaluation" was established (Fig. 1). Initially, the purity and optical absorption characteristics of the mineral powders were calibrated using XRD quantitative whole-rock analysis and UV–Vis-NIR spectrophotometry. Subsequently, disc-shaped specimens were subjected to laser irradiation under fixed parameters. The macro-scale damage morphology was reconstructed via 3D laser scanning, while the laws of mineral phase transition were analyzed through XRD testing of the ablation products. Based on the radial thermal damage gradient resulting from the Gaussian energy distribution of the laser beam, the irradiated region was categorized into the strongly affected zone, moderately affected zone, and weakly affected zone. Considering the distinct laser-induced surface responses of different minerals under identical irradiation conditions, a universal quantitative threshold was not adopted. Instead, the zoning was determined according to the radial variation in surface color together with the distance from the laser spot center, allowing a consistent characterization of laser-affected regions across minerals with different laser response characteristics. Finally, for the finely ground and polished cuboid specimens, the target mineral phases were identified and localized using SEM–EDS synergistic observation. On this basis, grid-based nano-indentation mapping was conducted to quantitatively analyze the micro-scale mechanical response characteristics of mineral grains across the three laser-affected zones.

2.2 Material Properties and Specimen Preparation

Granitic pegmatite deposits are typically found at depths ranging from 11 to 15 km, or even deeper (Fanka and Tadthai 2023). In China, they occur mainly within crustal-scale orogenic belts and are associated with the late-stage evolution of granitic magmatic systems. While the primary mineral assemblage (feldspar, quartz, and mica) is consistent with that of common granite, pegmatite is characterized by significantly coarser grain sizes. Accessory minerals, such as tourmaline, fluorite, and rare-metal minerals, constitute a key distinguishing feature relative to conventional granitic rocks (Dashtseren et al. 2025). Accordingly, four representative rock-forming minerals were selected for investigation in this study: fluorite (F), quartz (Q), tourmaline (T), and plagioclase (PL).

2.2.1 UV–Vis-NIR Spectroscopy

To quantify the energy coupling efficiency between the laser beam and the mineral constituents, the spectral responses of four rock-forming minerals were characterized using a PerkinElmer Lambda 750 spectrophotometer. As illustrated

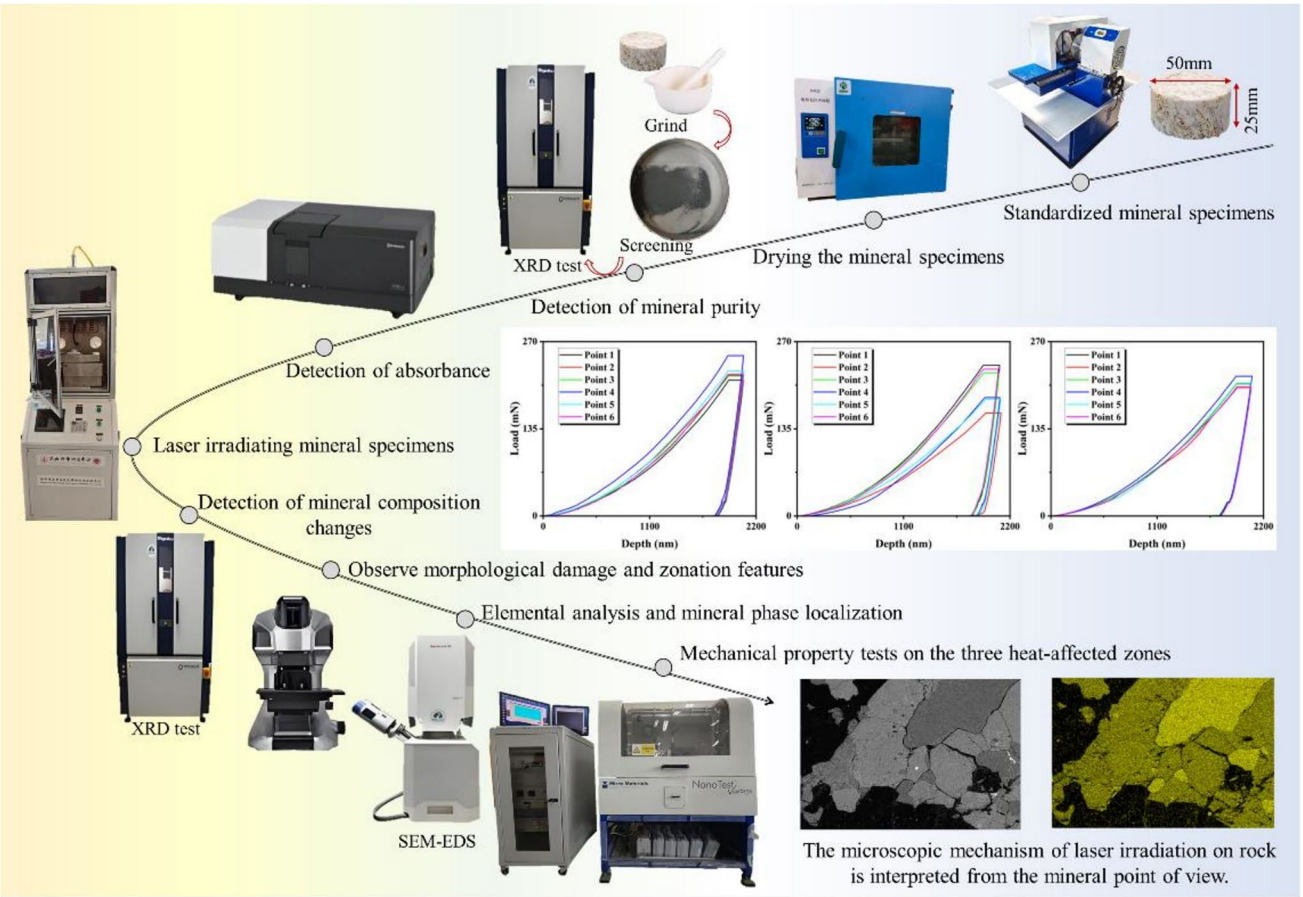


Fig. 1 Experimental workflow

in Fig. 2, the minerals exhibit pronouncedly differentiated absorbance at the 1080 nm laser operating wavelength, revealing a strong dependence of optical behavior on mineral species.

Specifically, T maintains an absorbance near unity, demonstrating near-perfect coupling with the near-infrared radiation, which implies efficient photothermal conversion. In contrast, PL shows a significant decline in absorbance, while Q and F manifest near-total optical transparency, with their absorbance values approaching zero. These findings are in strict accordance with the intrinsic optical properties of the constituent materials (Hahn 2014). This spectral disparity establishes the physical basis for the selective heating and subsequent heterogeneous thermal damage observed within the composite rock mass during laser irradiation.

2.2.2 XRD Purity Characterization of Mineral Constituents

To ensure the representativeness of the X-ray diffraction (XRD) measurements, both the pristine and laser-treated bulk minerals were pulverized using an agate mortar and sieved through a 200-mesh standard screen. This procedure yielded homogeneous powder specimens with a particle size distribution optimized for diffraction accuracy.

The pre-irradiation XRD analysis (Figs. 3 and 4) confirms the high purity and typicality of the selected mineral samples. Specifically, the F and Q specimens exhibit primary phase contents of 96.2% and 99.6%, respectively, estimated using the Reference Intensity Ratio (RIR) method in MDI Jade software. The T sample is predominantly composed of the T phase (99.1%), with only negligible traces of PL (0.9%). While the PL sample exhibits a more complex composition, it is dominated by feldspar-group minerals (38.3% K-feldspar and 26.6% PL, totaling 64.9%), complemented by Q (22.1%), calcite (5.4%), and minor clay impurities (7.5%). Given that the feldspar content exceeds 50%, the specimen satisfies the mineralogical requirements for this study.

2.3 Laser Experimental System and Parameters

To mitigate the thermal interference of moisture, all mineral specimens were preconditioned at 110°C for 48 h to

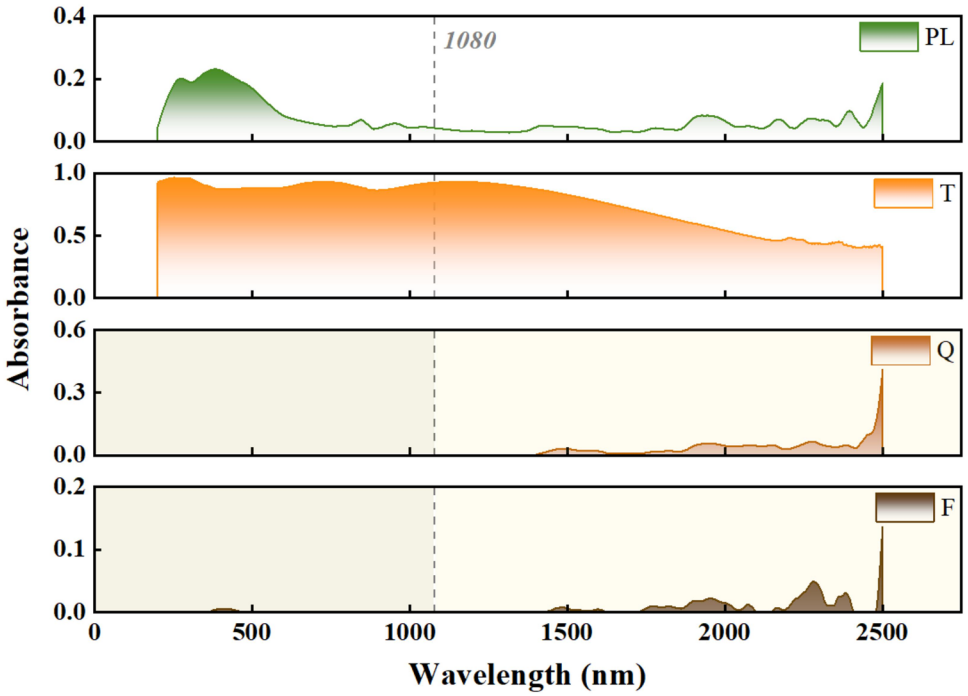


Fig. 2 UV–Vis-NIR absorbance spectra of the four-constituent rock-forming minerals

ensure the complete desorption of both adsorbed and interlayer water (Lang et al. 2017). The irradiation experiments were conducted using a custom-developed laser cracking system for deep hard rock. This apparatus integrates a 1080 nm fiber laser, a precision displacement stage, an auxiliary gas control unit, and a monitoring module. Specifically, an internal infrared thermograph (range: −20 ~ 925°C; resolution: 382 × 288 pixels) and an industrial camera were utilized for the in-situ capturing of surface temperature evolution and fracture initiation. To ensure statistical consistency, triplicate tests were performed for each mineral species. The laser parameters were fixed at a power of 3 kW, a duration of 30 s, and a beam spot diameter of 15 mm. The experimental configuration is schematically illustrated in Fig. 5.

2.4 Characterization Methods for Mineral Damage

2.4.1 Post-irradiation Mineralogical Characterization

The phase evolution of the laser-irradiated minerals was characterized using a Rigaku Ultima IV diffractometer (operated at 40 kV and 40 mA). To ensure high peak resolution, diffraction patterns were acquired in continuous scanning mode from 5° to 80° with a precision step size of 0.02°. Further refinement of the diffraction data was performed to semi-quantitatively evaluate the mineralogical transformations and variations in relative content.

2.4.2 3D Morphological Characterization of Laser-induced Damage

The surface morphology of the irradiated minerals was reconstructed using a KEYENCE VR-5000 wide-area 3D profilometer. A 12 times magnification was employed to ensure full coverage of the damage-affected zone. To maintain the high fidelity of the topographic data, the acquired 3D models were processed using the built-in VR Analyzer software of the KEYENCE VR-5000 system for datum plane correction and noise reduction.

2.4.3 SEM–EDS Based Mineral Phase Identification

The laser-irradiated mineral specimens were characterized using a Thermo Scientific Apreo 2C scanning electron microscope (SEM). Twelve experimental scenarios were defined based on a full factorial design involving four representative minerals (F, Q, T, and PL) and three laser-affected zones (strong, medium, and weak), resulting in $3 \times 4 = 12$ experimental conditions. By integrating elemental distribution mapping via EDS (conducted with a sampling interval of 5 μm) with the detection of representative mineral constituents, local mineral phases were identified and calibrated. This characterization provided a critical spatial reference for the subsequent zonal nano-indentation tests.

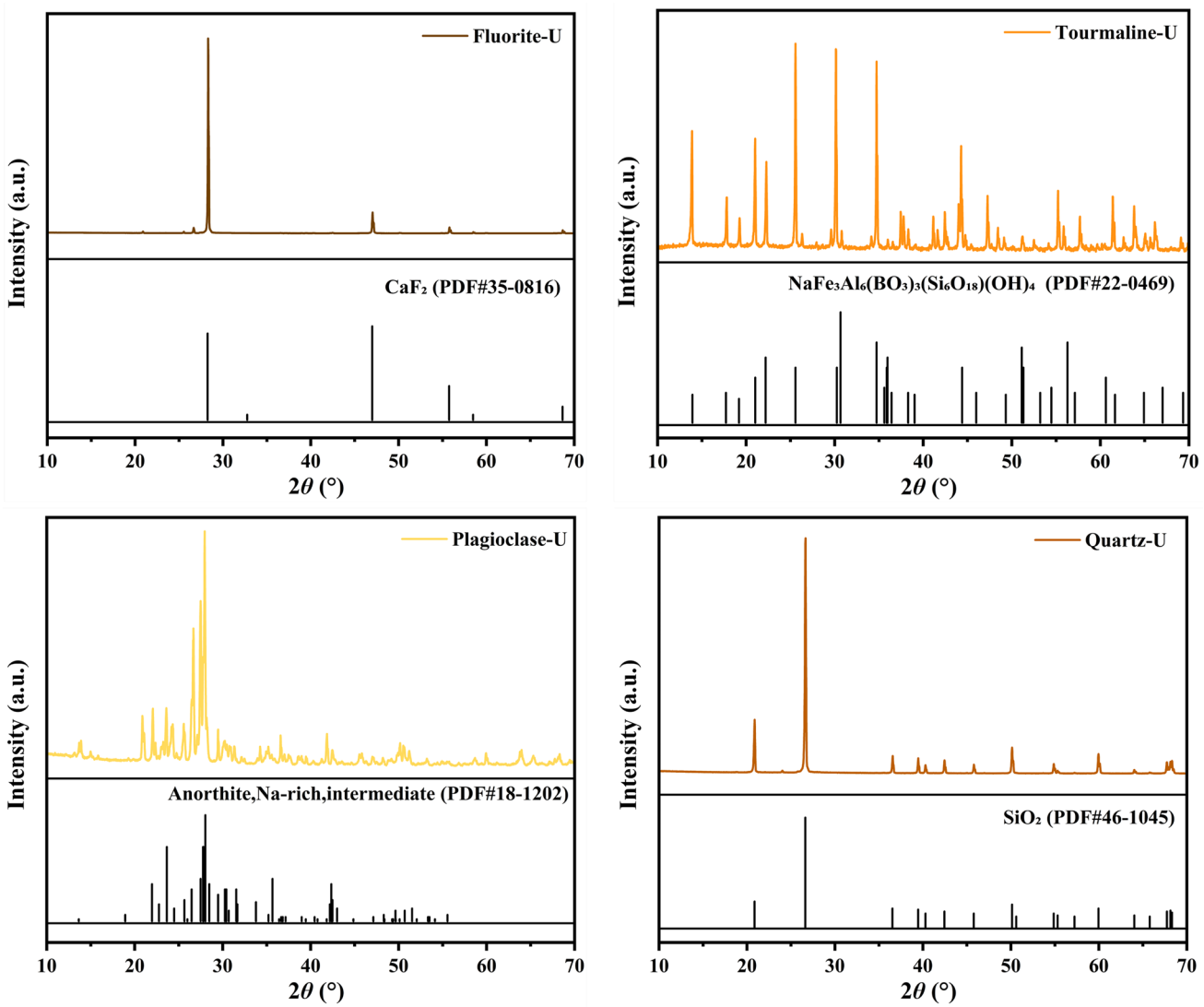


Fig. 3 XRD patterns of the four constituent minerals prior to laser irradiation

2.4.4 Zonal Micromechanical Characterization via Nano-indentation

The micromechanical properties of the mineral constituents were characterized using a NanoTest system (Micro Materials, UK) equipped with a Berkovich diamond indenter (three-sided pyramidal geometry), featuring a displacement resolution of 0.01 nm and a load resolution of 50 mN. To mitigate data scattering arising from material heterogeneity inherent in load-controlled testing, a depth-controlled mode was adopted (Shen et al. 2022). Following preliminary trials, the maximum indentation depth was optimized at 1000 nm to strike a balance between neutralizing surface roughness effects and minimizing thermal drift accumulation (Oliver and Pharr 2004). To ensure the precision of hardness (H) and Young's modulus (E) measurements, thermal drift was suppressed through two synergistic strategies: (i) a rapid testing protocol employing a loading/unloading rate of 2 mN/s to shorten the cycle and inhibit drift accumulation, and (ii) real-time compensation via the system's automated calibration procedure to dynamically correct micro-displacement deviations. These measures ensured that thermal drift interference remained consistently below the instrument's detection threshold.

Specimens with dimensions of approximately 15 mm × 15 mm × 10 mm were extracted from the strong, medium, and weak laser-affected zones. Following bilateral leveling, the samples were encapsulated in epoxy resin (Fig. 6) and subjected to a rigorous surface preparation protocol. This involved sequential grinding with high-mesh SiC papers (up to 3000 mesh) and fine polishing on a Struers polisher using 5.00 µm and 0.05 µm diamond suspensions. Anhydrous ethanol was

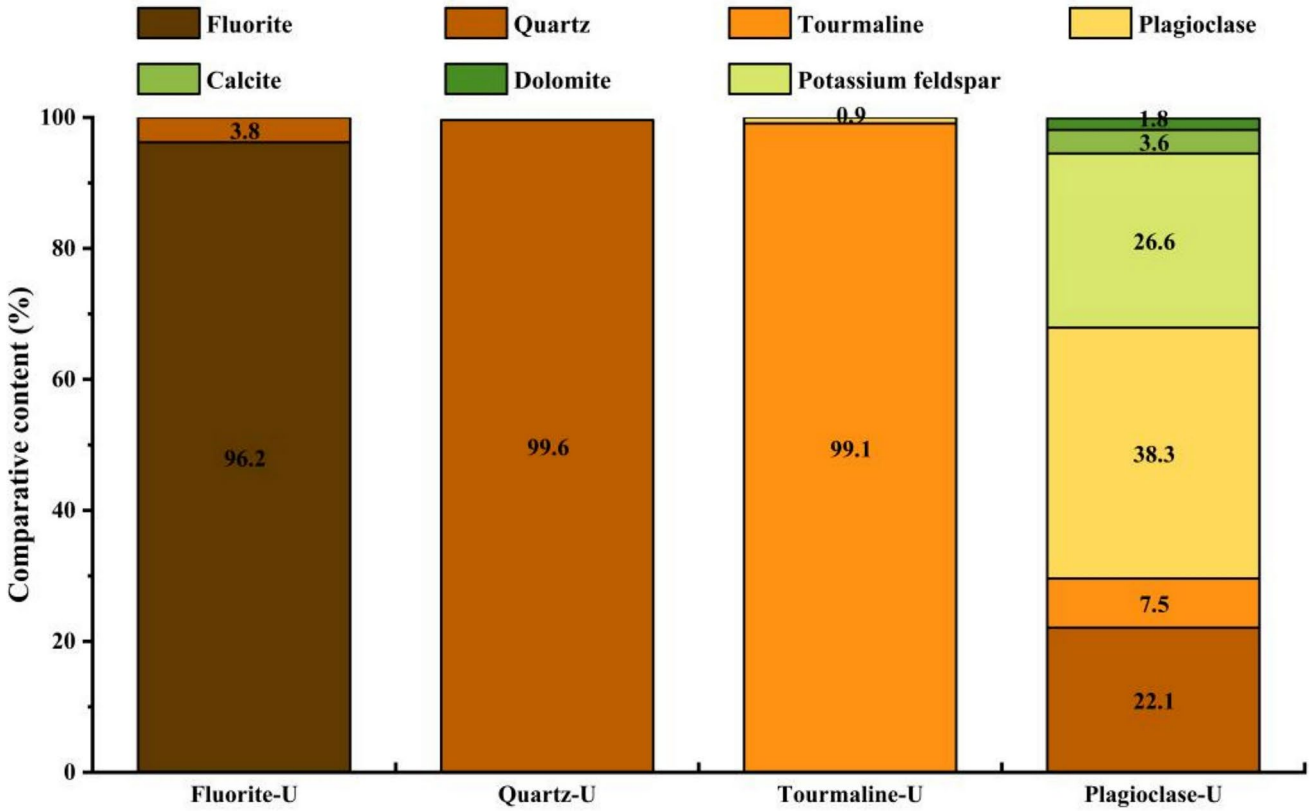


Fig. 4 Phase compositions and corresponding impurity concentrations of the mineral samples before laser treatment

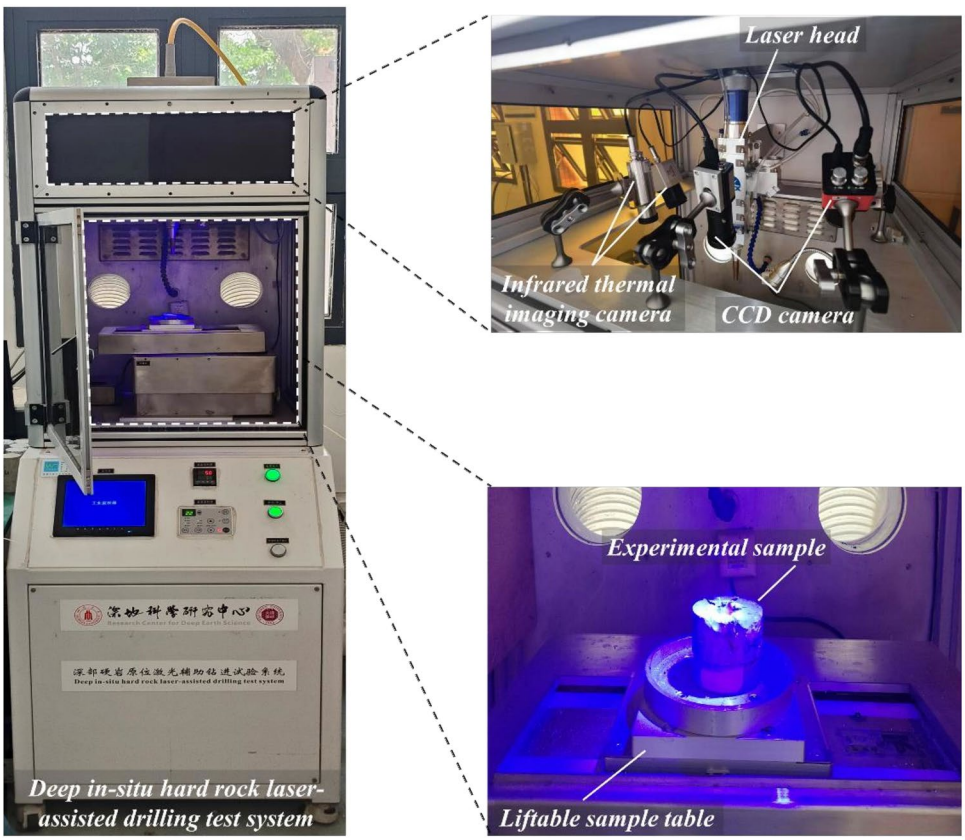


Fig. 5 Custom-developed 1080 nm high-power laser fracturing system

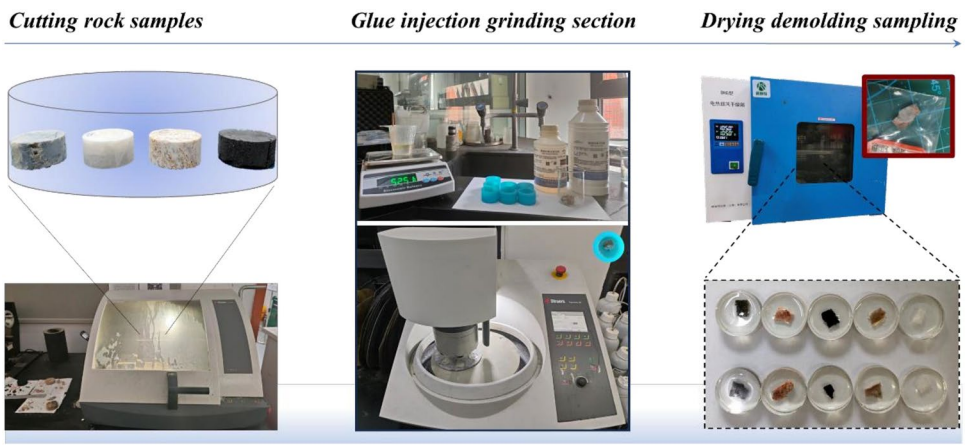


Fig. 6 Specimen preparation process for nano-indentation

utilized throughout for cleaning to ensure a contaminant-free surface, followed by drying at 50°C for 24 h. To account for mineralogical heterogeneity, a 2×3 grid indentation array was implemented across multiple locations (3 for pristine and 6 for irradiated samples). An indentation spacing of 20 μm was maintained to eliminate stress field superposition, as this distance significantly exceeded both the maximum penetration depth and the typical grain size. Prior to nano-indentation testing, indentation sites were selected using the built-in high-magnification optical imaging system of the nano-indentation instrument to ensure surface integrity and exclude structural defects. A total of 18 valid tests per pristine sample and 36 per irradiated sample were conducted to mitigate random errors.

3 Results

3.1 Comparative Analysis of Thermal Response Patterns

As the primary thermal driver, the spatial distribution of laser energy dictates the formation and spatiotemporal evolution of the temperature field within the mineral specimens. The infrared thermography data provides a critical experimental basis for decyphering energy transport mechanisms, bridging the gap between external energy input and internal heat conduction. Furthermore, the morphology and expansion dynamics of the observed high-temperature zones serve to validate the Gaussian beam model and provide a benchmark for 2D transient thermal modeling.

The energy density distribution of the laser beam is defined as follows (Galbusera et al. 2024):

$$q(x,y) = \frac{2P_0}{\pi r^2} \exp\left(-2\frac{\left(x-x_c\right)^2 + \left(y-y_c\right)^2}{r^2}\right) \quad (1)$$

where P_0 is the laser power (W); r is the beam radius (mm); $\left(x_c, y_c\right)$ and (x, y) represent the coordinates of the beam center and any given point within the heat-affected zone, respectively; and $q(x, y)$ denotes the local energy density (W/mm^2).

By substituting the laser parameters ($P_0 = 3000\,W$, $r = 7.5\,mm$) into the laser energy density equation, the maximum peak energy density under ideal conditions was calculated to be 33.9 W/mm^2 (Fig. 7). The spatial distribution characteristics of the theoretical energy density serve as the fundamental analytical benchmark for evaluating the subsequent actual laser-mineral interactions and thermal responses.

The infrared thermal imaging sequences (Figs. 8, 9, 10, 11, 12) reveal distinct thermal response patterns and energy dissipation mechanisms among the four rock-forming minerals under laser irradiation:

(1) During the initial irradiation stage ($t = 2.66 \sim 4.92$ s), F exhibited exceptionally high energy absorption efficiency. The focal temperature rapidly approached the detection upper limit (925°C), triggering instantaneous melting and localized vaporization. Distinct radial splashing observed in the thermal images indicates that the molten phase underwent dynamic migration driven by vapor pressure gradients. Over time ($t = 7.99 \sim 30.03$ s), the energy distribution evolved from localized rapid deposition to broad thermal diffusion. The intermittent hot spots observed during this stage are attributed to localized internal phase transitions and the escape of gaseous products.

(2) Q displayed a typical conduction-dominated mode, characterized by steady and continuous temperature rise throughout the process, with no evidence of melt splashing or plume ejection. Starting from $t = 4.88$ s, a prominent radial temperature gradient emerged at the center and expanded steadily outward. By $t = 30.03$ s, the heat flow had essentially covered the entire specimen surface. Such stable temperature field evolution reveals that Q possesses high thermal stability and a

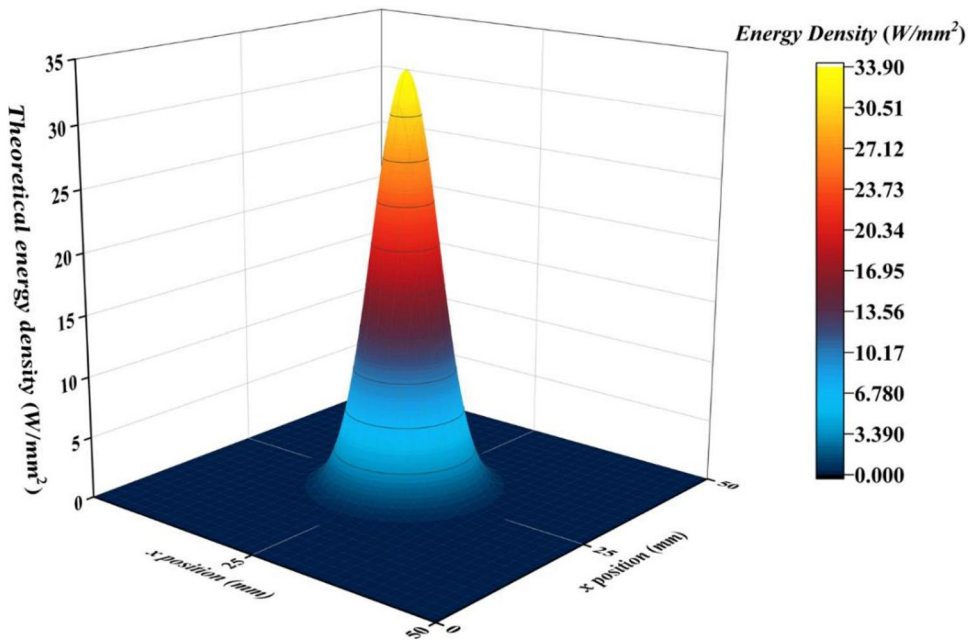


Fig. 7 Theoretical energy density distribution based on the Gaussian beam model

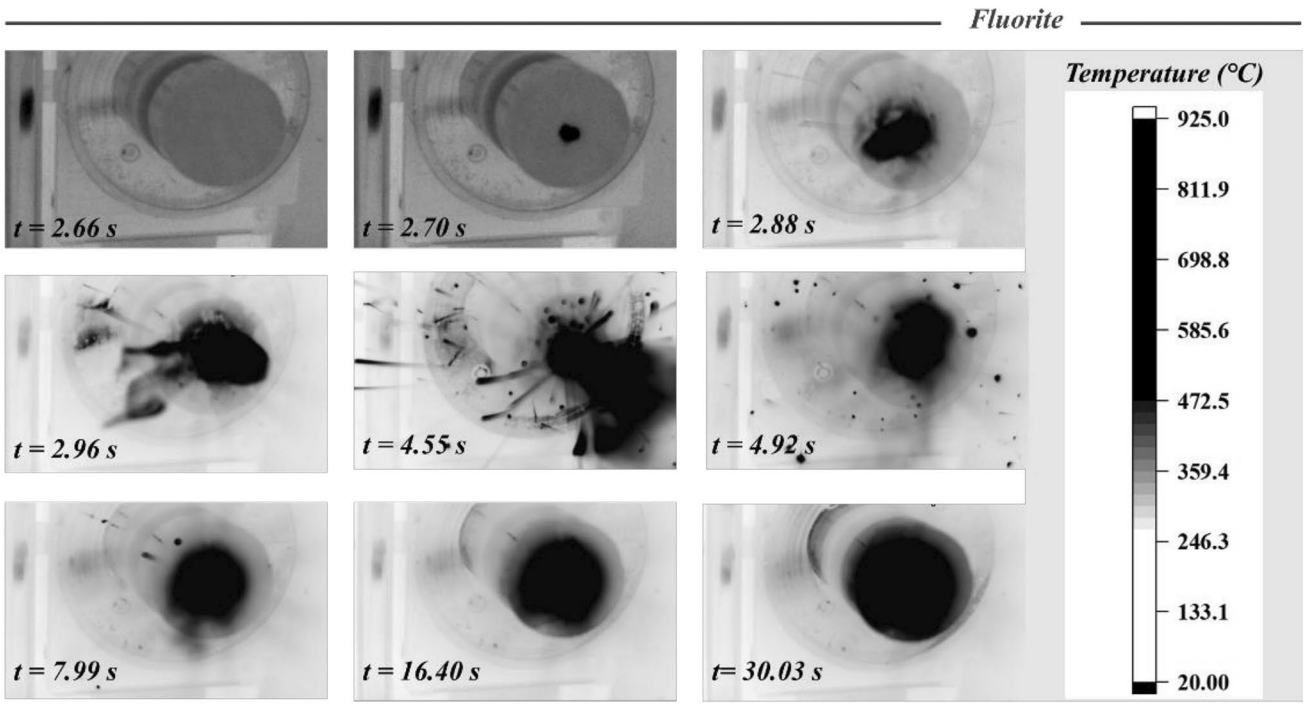


Fig. 8 Spatiotemporal evolution of infrared thermal imaging on F under laser irradiation. t represents irradiation time (s). The color bar indicates temperature (°C)

singular thermophysical response under 1080 nm laser irradiation.

(3) The thermal evolution of T was accompanied by violent physicochemical transformations. Around $t = 4.44$ s, intense smoke-like plumes erupted from localized high-temperature zones, signifying its pronounced laser-thermal sensitivity and induced vaporization-decomposition. As energy input continued, the high-temperature region expanded radially toward the boundaries. In the late stage ($t = 28.85 \sim 30.03$ s), high-energy molten particles were observed escaping from the matrix, while

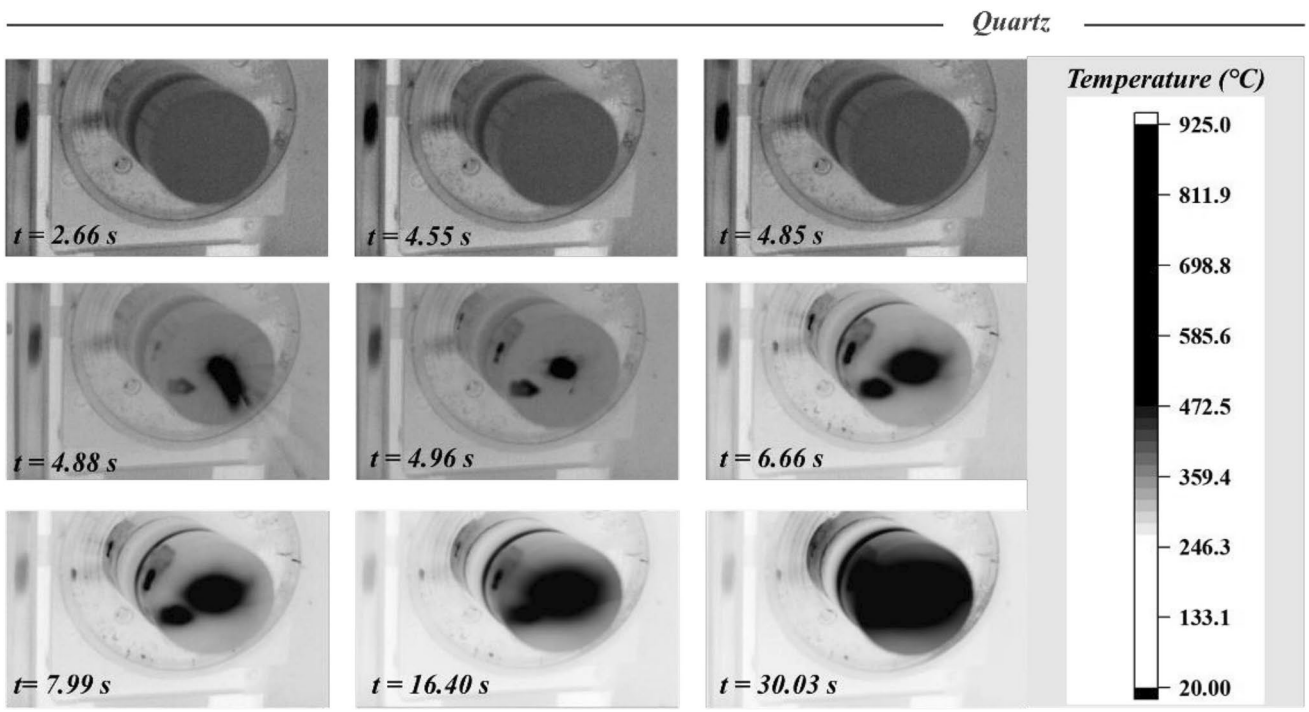


Fig. 9 Spatiotemporal evolution of infrared thermal imaging on Q under laser irradiation. t represents irradiation time (s). The color bar indicates temperature (°C)

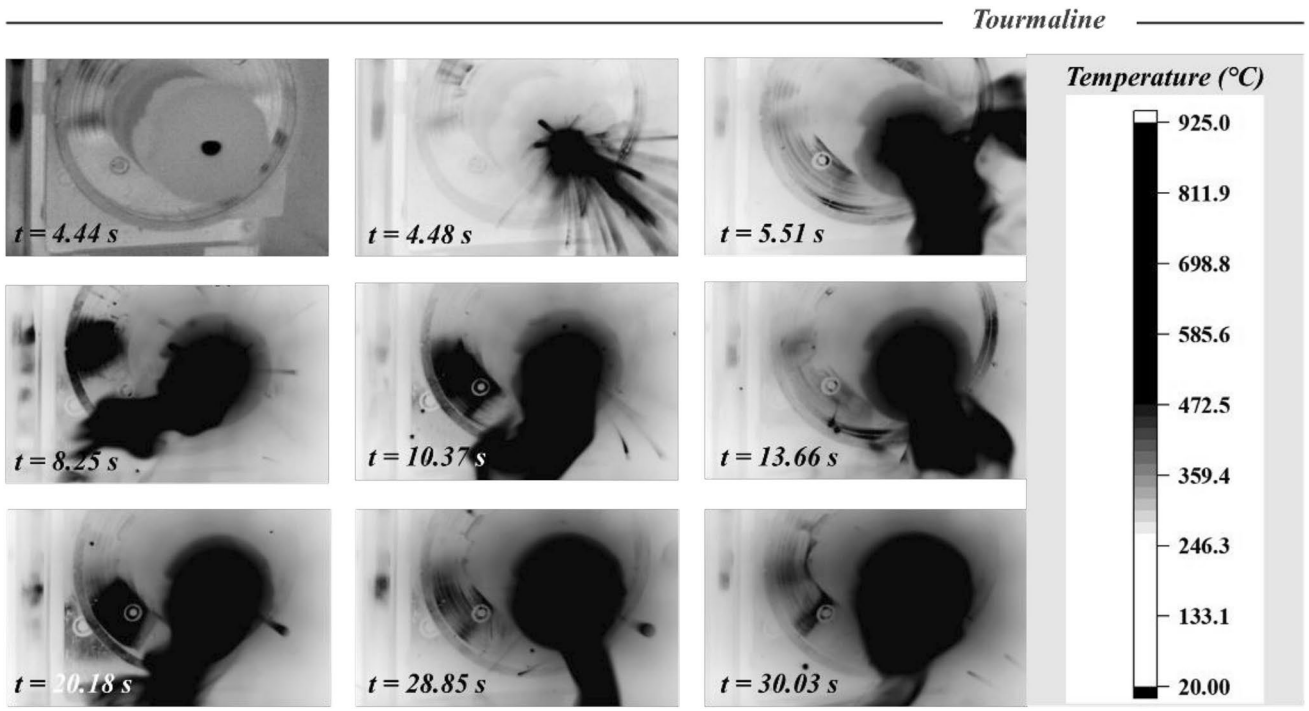


Fig. 10 Spatiotemporal evolution of infrared thermal imaging on T under laser irradiation. t represents irradiation time (s). The color bar indicates temperature (°C)

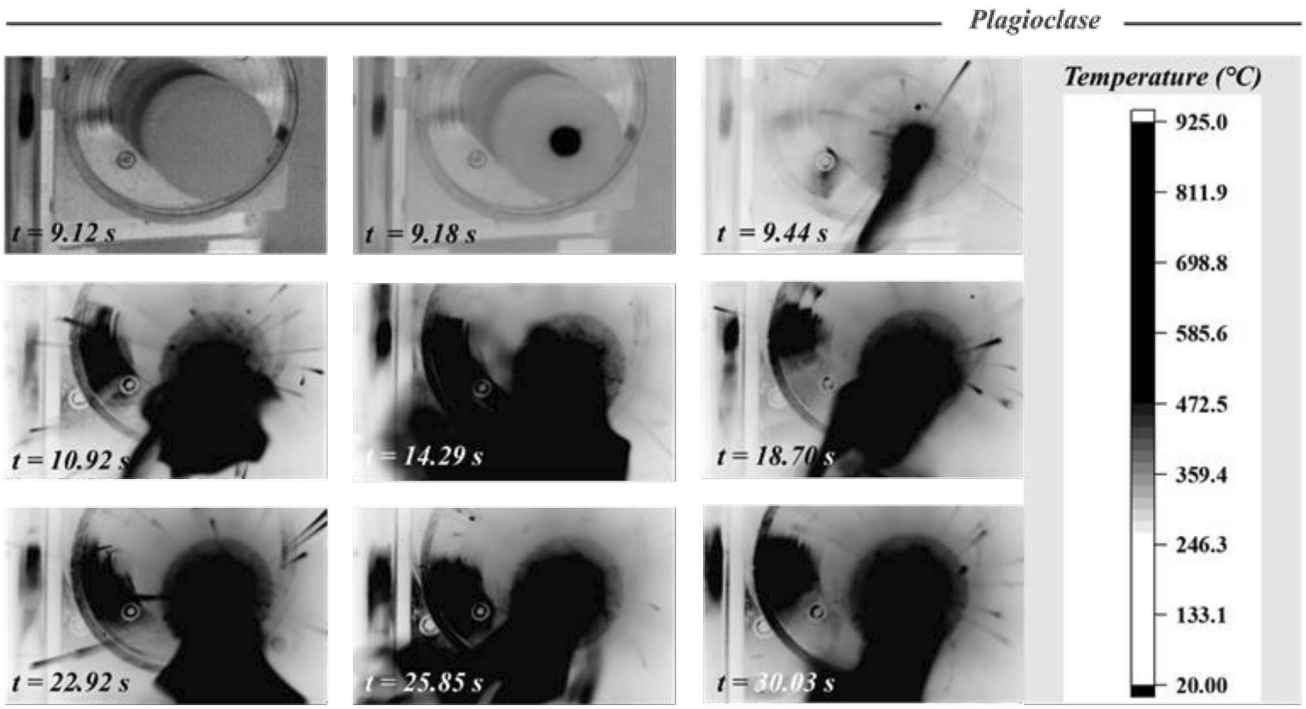


Fig. 11 Spatiotemporal evolution of infrared thermal imaging on PL under laser irradiation. t represents irradiation time (s). The color bar indicates temperature (°C)

the bulk material maintained a high-temperature state (600 ~ 900°C).

(4) This mineral exhibited the most complex failure characteristics. Localized hot spots first emerged at $t = 9.12 \sim 9.18$ s, subsequently evolving into violent thermal decomposition jets between $t = 10.92 \sim 14.29$ s, accompanied by the high-velocity ejection of volatiles and micro-scale melts. By $t = 30.03$ s, the diffused plumes caused the thermal boundaries to become blurred, reflecting the heterogeneous thermal damage behavior of PL's complex mineralogical composition under laser irradiation.

3.2 Zonal Damage Characteristics of Rock-forming Minerals Post-Laser Irradiation

Morphological analysis (Figs. 13 and 14) reveals that the four minerals exhibit distinct damage modes, characterized by a typical radial gradient that attenuates from the spot center to the periphery. Based on the radial damage evolution, the laser-affected area was divided into weak, medium, and strong zones (Fig. 14). For minerals with distinct zonation, the boundaries were determined based on radial morphological variations. For minerals with less pronounced zonation, the corresponding radial regions identified in minerals with distinct zonation were referenced for classification to reduce the influence of zonation differences on subsequent mechanical analysis.

(1) F exhibits a well-defined three-stage damage zoning. The highly affected zone (center) features a distinct depression covered by white colloidal aggregates, indicating pronounced melt-flow trajectories. The moderately affected zone (transition zone) is populated by micron-sized vitreous microspheres, which represent non-equilibrium phase products resulting from the rapid condensation of vaporized material. In contrast, the weakly affected zone (periphery) retains original polishing striations but exhibits dense thermal-stress-induced micro fissures at the crystal-matrix interfaces, revealing heterogeneous structural weakening.

(2) Q demonstrates exceptional laser resistance with high morphological continuity. Damage is confined to the focal center, appearing as micro-spots with high optical reflectivity. This abrupt change in optical properties suggests localized amorphization induced by transient high temperatures. This low thermal sensitivity is intrinsically linked to its high melting point and stable 3D network crystalline structure.

(3) T exhibits the most intense regional differentiation. The central zone is dominated by a flat melt pool covered with oil-luster black vitreous phases. The surrounding bright-white reflective ring is attributed to phase tran-

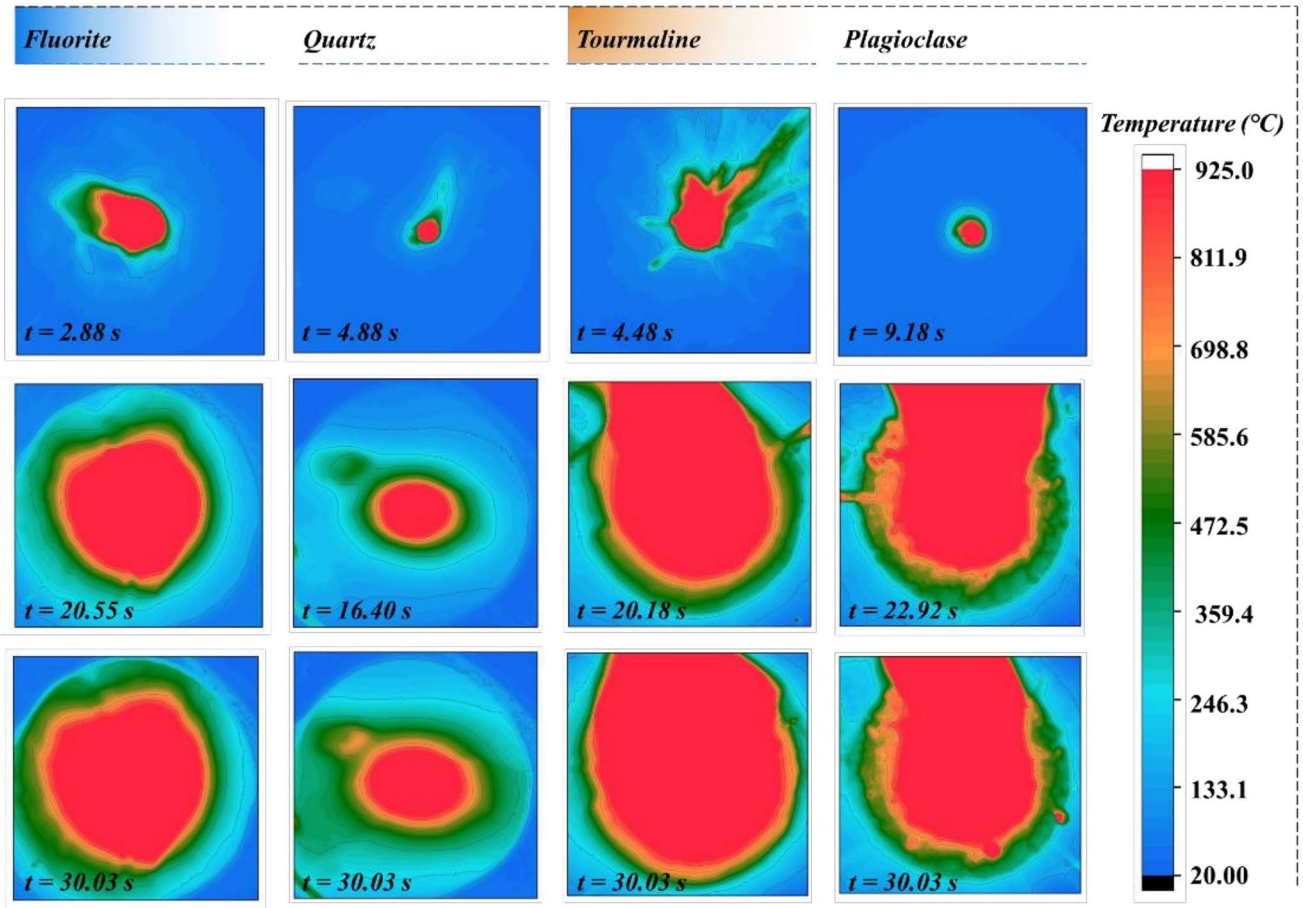


Fig. 12 Differentiated surface temperature field evolution characteristics of four rock-forming minerals under laser irradiation. t represents irradiation time (s). The color bar indicates temperature (°C)

sitions of high-temperature re-condensates. The outermost matrix shows typical brittle failure, characterized by a dense microcrack network originating from the center that occurs as a result of the synergistic effects of explosive volatile release and extreme thermal shock.

(4) PL displays a continuous transition from a central transparent protrusion to a re-condensed ring and a peripheral weak-impact zone, aligning well with the Gaussian energy distribution. The central transparent vitreous protrusion is a new phase formed by rapid cooling post-melting. This is followed by an irregular vitreous re-condensation ring. The peripheral zone exhibits a light orange matrix with dispersed microcracks and interspersed impurities.

Furthermore, the 3D profile reconstruction and the resulting color-coded elevation maps (Fig. 15) provide a quantitative characterization of the spatial damage features of the four minerals under laser irradiation:

(1) F exhibits pronounced localized thermal ablation. A distinct ablation pit forms at the beam center, where the elevation drops precipitously from 0.44 mm to -3.40 mm, resulting in a maximum ablation depth of 3.84 mm. The sharp tonal contrast in the elevation map (ranging from deep blue at the center to yellowish-brown at the periphery) visually illustrates the rapid attenuation of the thermal response with increasing radial distance.

(2) In contrast, Q demonstrates superior thermal stability with extremely limited morphological alteration. The overall elevation fluctuation (−0.24–−0.76 mm) is significantly lower than that of the other minerals, with a maximum differential of only 0.52 mm. Only localized micro-pits are observed in the central region; this micro-scale selective etching is likely attributable to differential energy absorption induced by trace impurities within the mineral matrix.

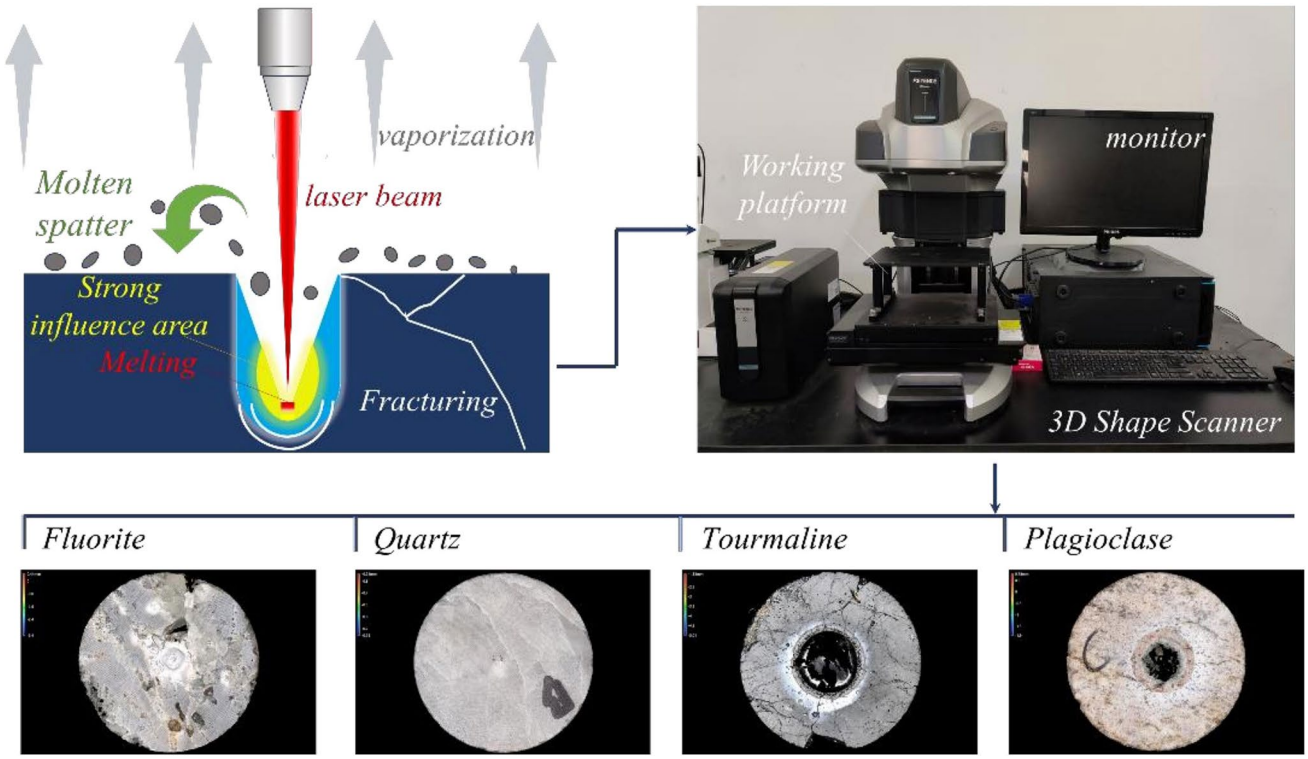


Fig. 13 Surface morphological scanning of rock-forming minerals following laser irradiation

(3) T displays significant topographic differentiation. The elevation plunges from −1.83 mm at the edges to −5.01 mm at the center, creating a relief of approximately 3.18 mm. Beyond the red–orange high-value zones formed by central melting, a network of radially distributed micro-cracks is clearly visible in the transition zone, indicating that T underwent brittle fracturing under the extreme thermal field. The localized data gaps in the central region are primarily due to scanning signal distortion caused by specular reflection from the molten products.

(4) The damage characteristics of PL align closely with the Gaussian distribution of the laser energy. The morphology exhibits a continuous spatial elevation gradient (maximum relief of 3.64 mm) with marked axisymmetric. The central deep blue zone corresponds to an area of intense melting and evaporation, while the yellow-green gradient in the transition zone reflects energy attenuation, with the orange-red peripheral areas approaching the original surface. This spatial distribution indicates that the morphological evolution is primarily driven by the radial thermal gradient field induced by the spatial profile of the laser beam.

3.3 Elemental Composition and Identification of Rock-Forming Minerals

The sample surfaces were characterized using SEM at 500 × magnification, coupled with EDS. Typical rock-forming minerals were identified based on their representative elemental distributions (Fig. 16) to support spatial correlation during subsequent nanoindentation testing via the integrated optical microscope.

As shown in Fig. 16, the elemental mapping provides a clear visualization of the spatial distribution of distinct mineral phases. Specifically, regions enriched in F and Ca were identified as F; areas predominantly containing Si without Al corresponded to Q; regions characterized by Mg and Fe enrichment without Na were assigned to T; and zones containing Na or K were classified as PL. It should be noted that the primary objective of the EDS analysis was the qualitative identification and spatial mapping of mineral phases rather than rigorous quantitative elemental analysis or the investigation of elemental migration and rearrangement behaviors.

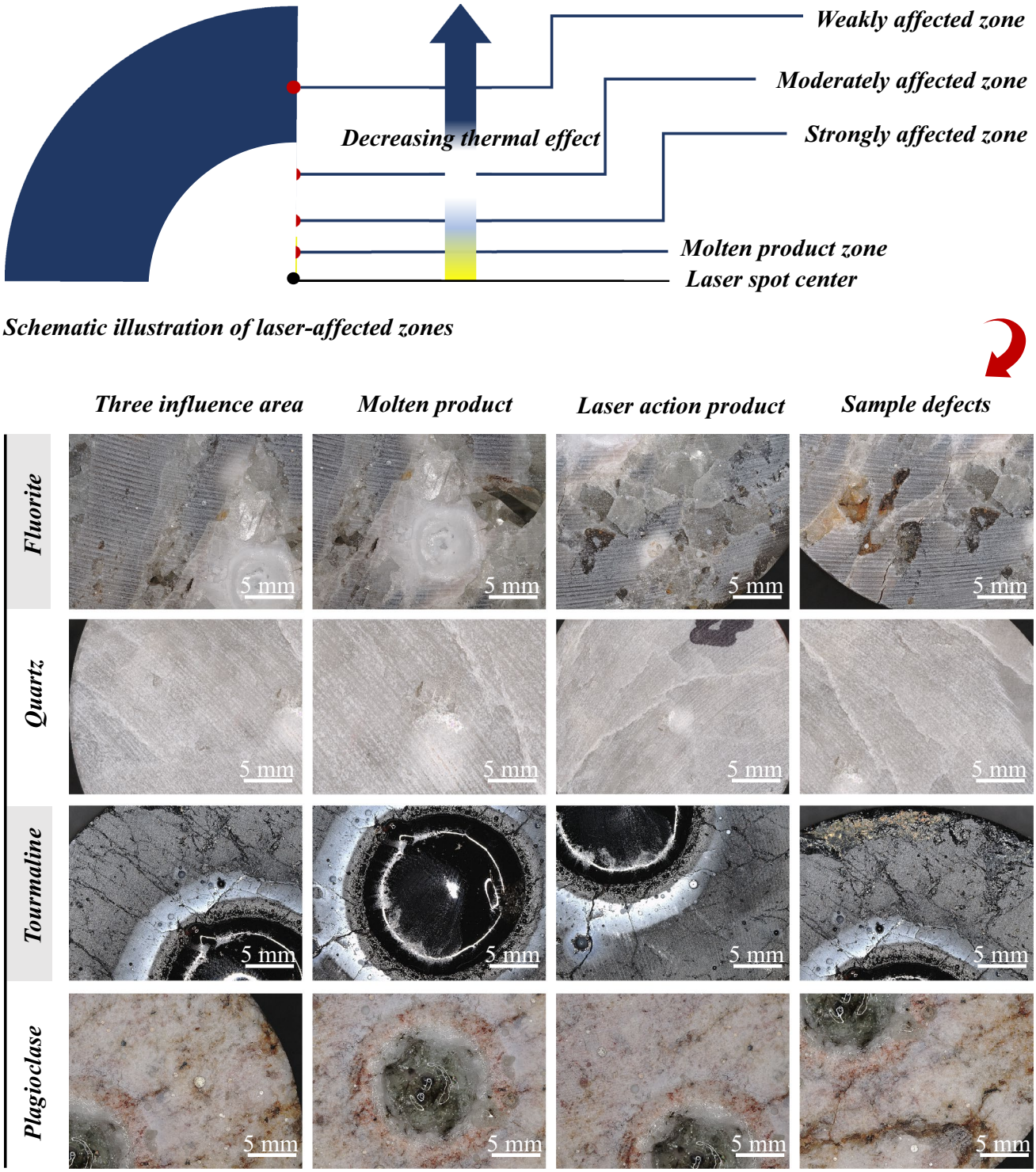


Fig. 14 Morphological evolution and damage zonation of rock-forming mineral surfaces under laser irradiation

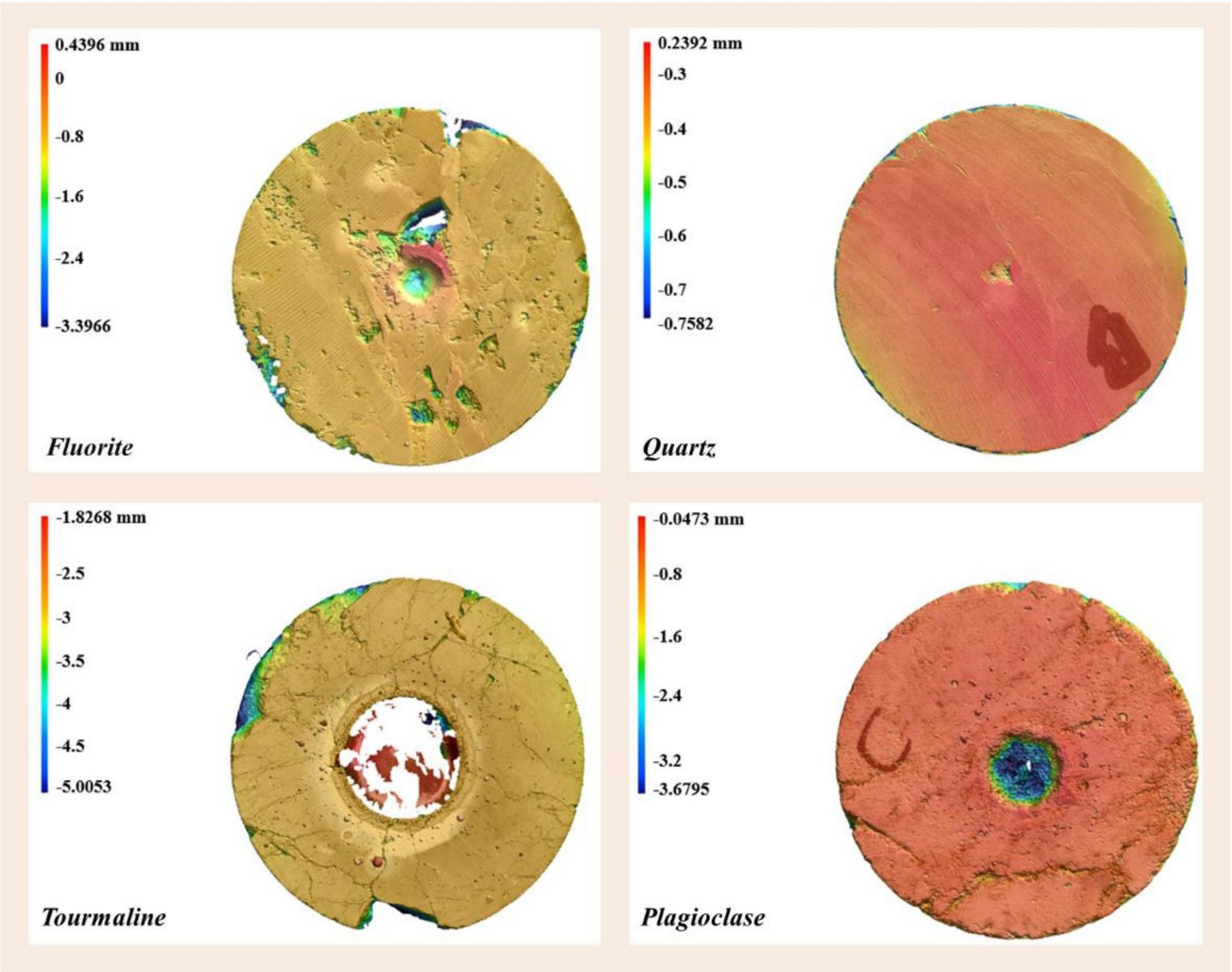


Fig. 15 Three-dimensional scanning color-coded elevation maps of mineral surfaces after laser irradiation

3.4 Laser-Induced Zonal Micromechanical Properties of Rock-Forming Minerals

3.4.1 Principles of Nanoindentation Testing

Nanoindentation technology applies micro-loads ($nN \sim mN$) to the material surface and records the load–displacement curves in real-time, from which key parameters such as hardness and elastic modulus are calculated based on contact mechanics theory. Figure 17a illustrates the indentation profile, where h_c is the contact depth corresponding to the peak load (mm), h_s is the surface sink-in depth (mm), and h_p is the residual indentation depth after unloading (mm). Figure 17b shows a typical load-depth curve, where $F_{\max}$ is the peak load (mN) and $h_{\max}$ is the maximum indentation depth (mm).

3.4.2 Processing of Load-Depth Curves Based on the Oliver-Pharr Method

The contact stiffness (S) of the indentation region is determined by differentiating the initial portion of the unloading curve at the maximum penetration depth:

$$S = \frac{dF_n}{dh}|_{h=h_{\max}} \tag{2}$$

The contact depth (h_c) is calculated as:

$$h_c = h_{\max} - \varepsilon\left(h_{\max} - h_r\right) \tag{3}$$

The projected contact area (A_c) of the indenter is intrinsically linked to the contact depth (h_c). For a Berkovich

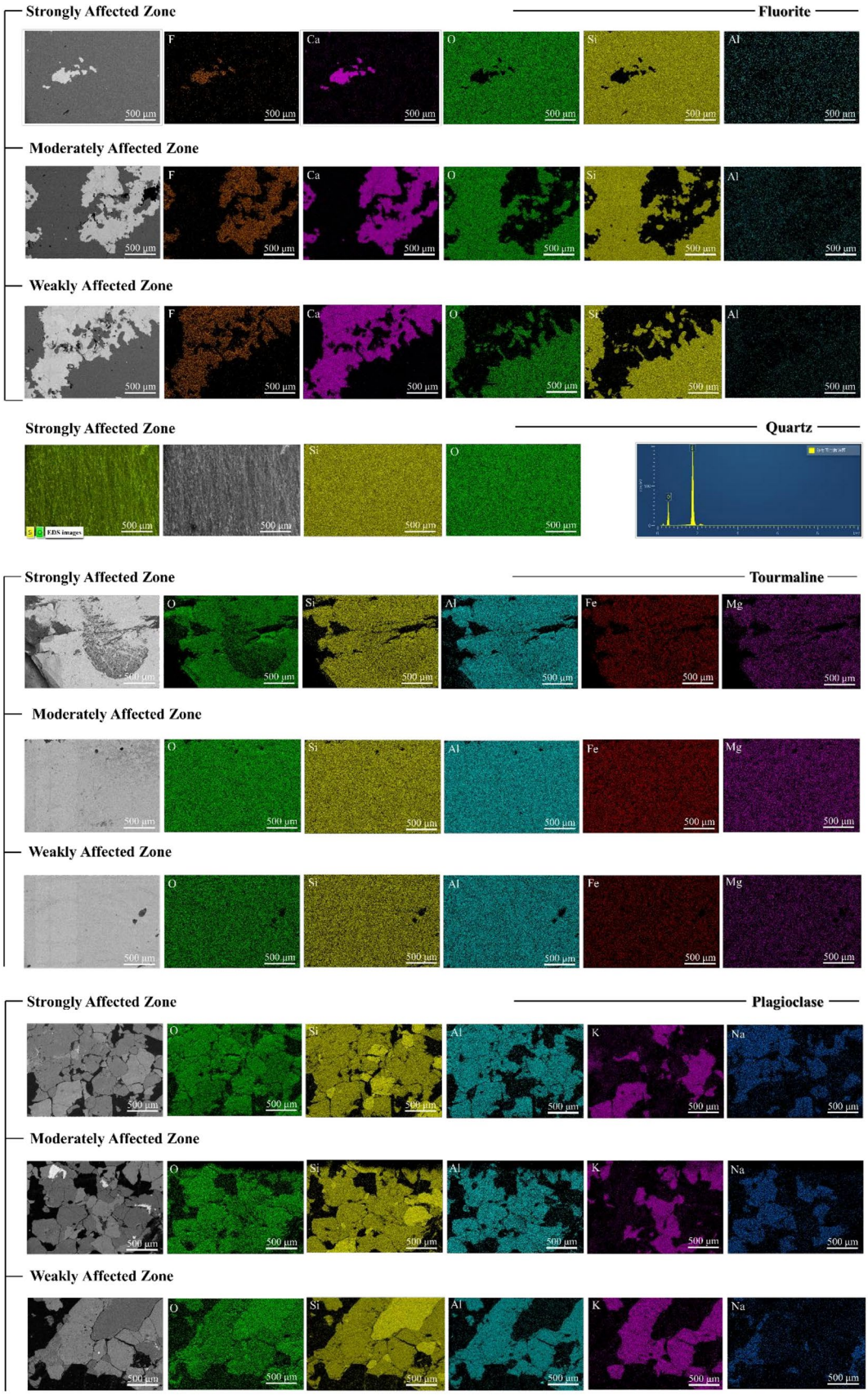


Fig. 16 Elemental distributions within three distinct heat-affected zones for rock-forming minerals after laser irradiation

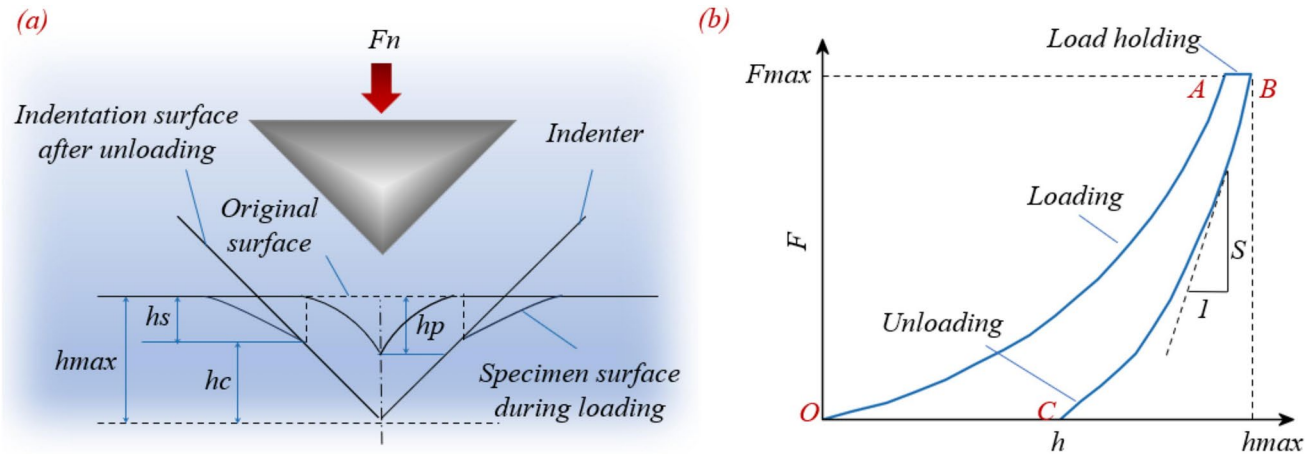


Fig. 17 Schematic illustration of indentation morphology and typical load-depth curve

indenter, this relationship is expressed as (Oliver and Pharr 1992):

$$A_c = 24.5h_c^2 + C_1h_c^{1/2} + ... + C_1h_c^{1/128} \tag{4}$$

According to the Oliver-Pharr method, the reduced modulus (E_r) and hardness (H) of the minerals are derived as follows:

$$E_r = \frac{S\sqrt{\pi}}{2\eta\sqrt{A_c}} \tag{5}$$

$$H = \frac{F_{\max}}{A_c} \tag{6}$$

where F_n is the indentation load (mN), h is the corresponding depth (mm), h_c is the contact depth, $h_{\max}$ is the maximum displacement, h_r is the residual displacement, ε and η are dimensionless constants related to the indenter geometry. In this study, a diamond Berkovich indenter was employed with $\varepsilon = 0.75$, $\eta = 1.034$, and $A_c = 24.56h_c^2$ (Oliver and Pharr 1992).

3.4.3 Influence of Laser Irradiation on Load-Depth Curves

The load-depth curve serves as a primary indicator for analyzing the evolution of material mechanical behavior. In this study, the spatial distribution of rock-forming minerals was identified via SEM–EDS, and the indentation grids were precisely localized using the integrated optical microscope of the nanoindentation system. Figure 18 illustrates the load-depth curves for the four minerals across three distinct heat-affected zones (strong, medium, and weak), comprising 72 tests (6 replicates per condition). All tests were conducted in displacement-control mode with a maximum depth of 2000 nm. Under constant depth conditions, the mineral hardness is directly correlated with the maximum load (Eq. 6). The results indicate that the load-depth responses vary significantly across the different heat-affected zones, reflecting a zonal mechanical response sensitive to thermal intensity.

Notably, the load-depth curves of the harder minerals (F and Q) showed high consistency, whereas T and PL exhibited increasing scatter (dispersion) as thermal influence intensified. This phenomenon aligns with observations by Chavoshi and Xu (2018) and may stem from the activation of localized defects or intensified inhomogeneous plastic deformation. PL exhibited the highest dispersion, reflecting significant mechanical heterogeneity linked to its multiphase composition and anisotropic crystalline structure. Furthermore, pop-in events were frequently observed in the loading segments of T, signifying abrupt localized plastic instability or transient micro-crack propagation. With intensified laser influence, the increased frequency of pop-ins and larger residual depths in T indicate a decline in the material's capacity to resist plastic deformation and inhibit crack growth, highlighting a trend of microstructural instability induced by laser treatment.

3.4.4 Influence of Laser Irradiation on Hardness and Reduced Modulus

The hardness of rock minerals represents their resistance to localized penetration and damage, while the elastic modulus characterizes their capacity to resist overall elastic deformation. According to the Oliver-Pharr method, since a constant indentation depth was maintained for all nanoindentation tests in this study, the projected contact area can be regarded as a constant. Consequently, the hardness is directly proportional to the maximum load. This implies that the evolution of hardness under laser treatment mirrors the developmental trends of the peak load observed in the load-depth curves. A

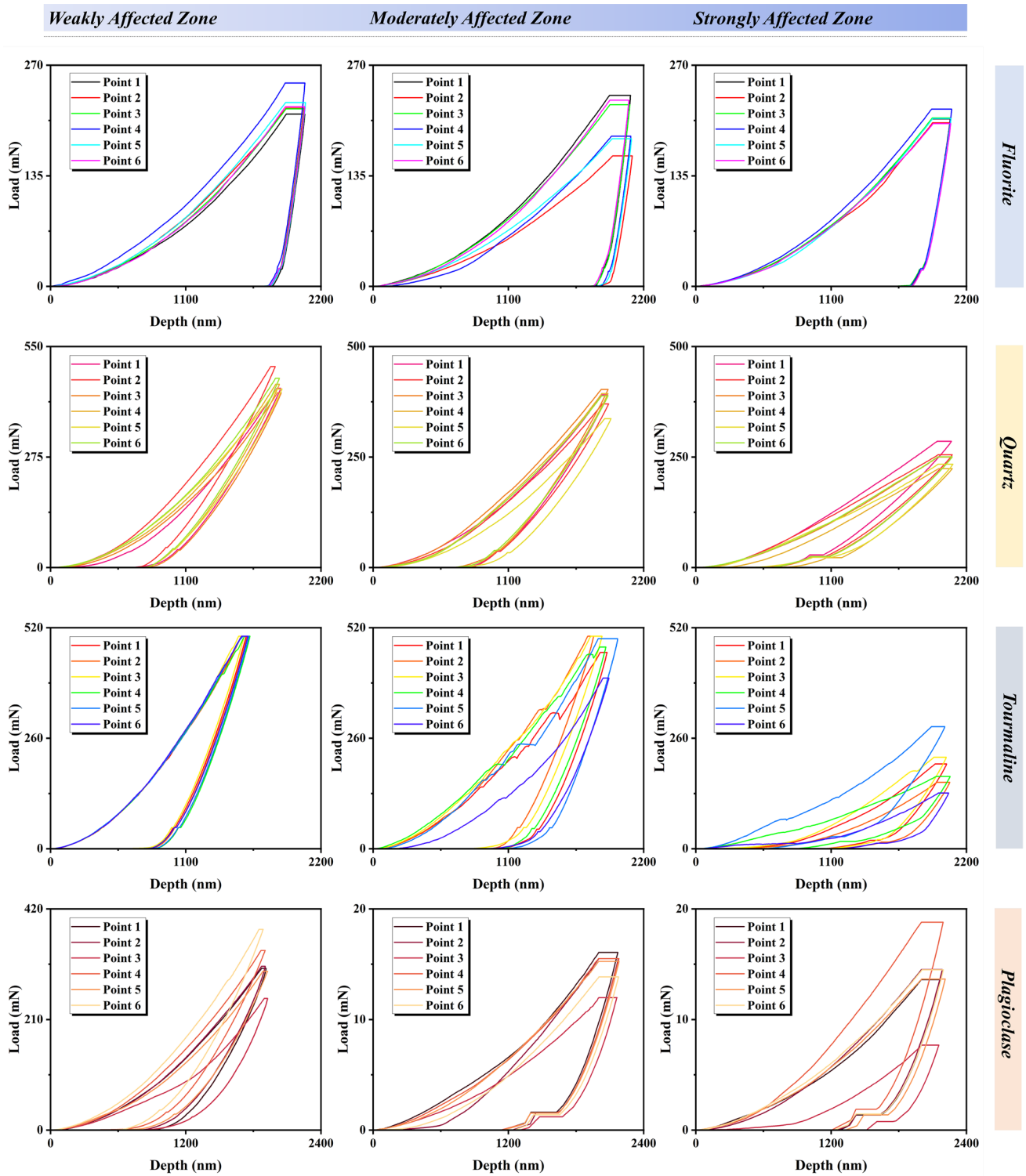


Fig. 18 Nanoindentation test results of rock-forming minerals across three distinct heat-affected zones following laser irradiation

detailed discussion regarding the evolution of the maximum load will be presented in the Discussion section.

In the weakly affected zone, the thermal perturbation of the laser is negligible, with mineral hardness remaining close to its intrinsic value. Table 1 indicates that the pristine hardness of Q and T is relatively high (10.29 GPa and 13.00 GPa, respectively), whereas PL (5.92 GPa) and F (2.39 GPa) exhibit lower initial resistance (The reduction analysis in this

Table 1 Hardness values of minerals across different heat-affected zones (GPa)

Zone	F			PL			Q			T		
	W	M	S	W	M	S	W	M	S	W	M	S
Point-1	2.25	2.42	2.22	5.42	0.18	0.15	9.73	8.24	5.87	13.32	7.51	2.62
Point-2	2.35	1.6	2.16	5.72	0.18	0.16	12.31	7.78	4.99	13.12	9.96	1.87
Point-3	2.33	2.32	2.23	3.85	0.13	0.08	9.22	8.94	4.76	13.7	9.17	2.86
Point-4	2.67	1.87	2.34	6.85	0.18	0.21	10.21	8.74	4.05	12.47	7.98	2.23
Point-5	2.39	1.84	2.21	5.52	0.17	0.14	9.74	6.54	4.03	12.59	7.28	4.34
Point-6	2.35	2.39	2.14	8.16	0.16	0.16	10.53	8.58	4.77	12.82	6.21	1.49

study is based on mean values). With intensified laser influence, all four minerals display varying degrees of weakening: Q decreases to 4.75 GPa in the strong HAZ, T drops to 2.57 GPa, and PL undergoes catastrophic structural degradation, plummeting from 5.92 GPa to 0.15 GPa. In contrast, F demonstrates superior mechanical stability, fluctuating only within the range of 2.07 ~ 2.39 GPa. To quantitatively evaluate the laser-induced weakening effect, the hardness reduction rates across different zones were calculated. From the weak to the strong HAZ, PL and T are most sensitive to laser radiation, with maximum hardness reduction rates exceeding 80%, followed by Q (> 50%), whereas F exhibits the minimum degree of weakening with a reduction of less than 15%.

Statistical analysis (Fig. 19) indicates that PL exhibits significant dispersion in the weak HAZ, which is attributed to its complex mineral composition and intrinsic structural heterogeneity. High-purity Q also shows pronounced data fluctuations; given that infrared monitoring indicates a minimal thermal response for Q, this anomaly is hypothesized to stem from localized perturbation damage induced during sample preparation (e.g., cutting and polishing). Deviations in mechanical response occur when indentation sites coincide with these preparation-induced defect zones. Furthermore, the dispersion of mechanical parameters for T increases further after laser treatment, which is ascribed to the proliferation of thermally induced micro-cracks and pores. This inference is highly consistent with the morphological evolution observed in Sect. 3.2, providing microstructural evidence for the heterogenization of mechanical responses. In contrast, the hardness values of Q and PL become more concentrated after laser irradiation, exhibiting reduced data dispersion. This behavior suggests that their relatively high intrinsic strength and structural homogeneity reduce the sensitivity of their mechanical properties to laser-induced thermal damage.

Statistical analysis by Yan et al. (2022) identified a linear correlation between the nanoscale hardness of minerals and

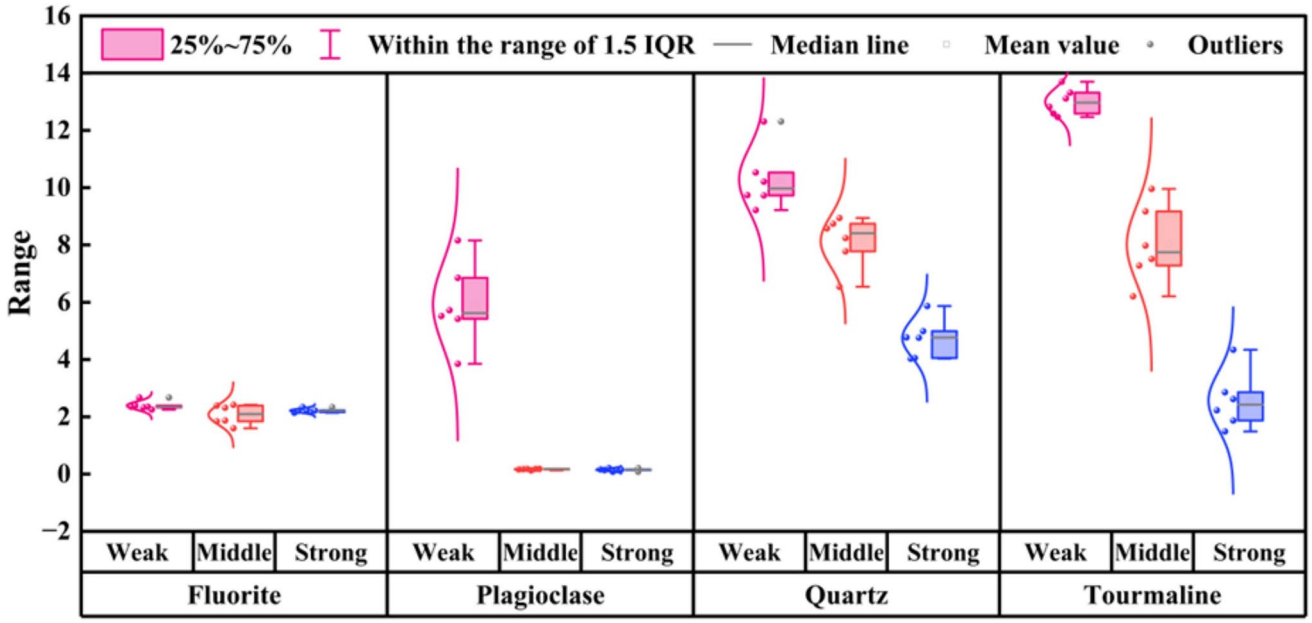


Fig. 19 Statistical distribution of hardness for rock-forming minerals across three laser-induced heat-affected zones. IQR represents the interquartile range

Table 2 Reduced modulus values of minerals across different heat-affected zones (GPa)

Zone	F			PL			Q			T		
	W	M	S	W	M	S	W	M	S	W	M	S
Point-1	110.77	125.69	96.06	64.97	3.89	3.52	81.86	68.88	39.37	143.59	115.01	51.63
Point-2	115.68	97.53	95.1	63.43	3.77	3.53	91.45	64.65	35.8	142.81	121.31	47.69
Point-3	112.31	117.71	95.26	64.6	3.41	2.78	81.18	68.03	36.07	145.76	114.19	53.85
Point-4	123.28	111.4	96.65	63.29	3.72	4.18	83.82	66.61	33.23	143.44	114.16	39.94
Point-5	115.89	105.73	96.73	58.93	3.67	3.53	79.64	60.78	35.83	144.19	114.87	57.87
Point-6	112.94	122.51	95.54	69.23	3.49	3.55	86.34	65.92	35.83	144.62	105.15	51.7

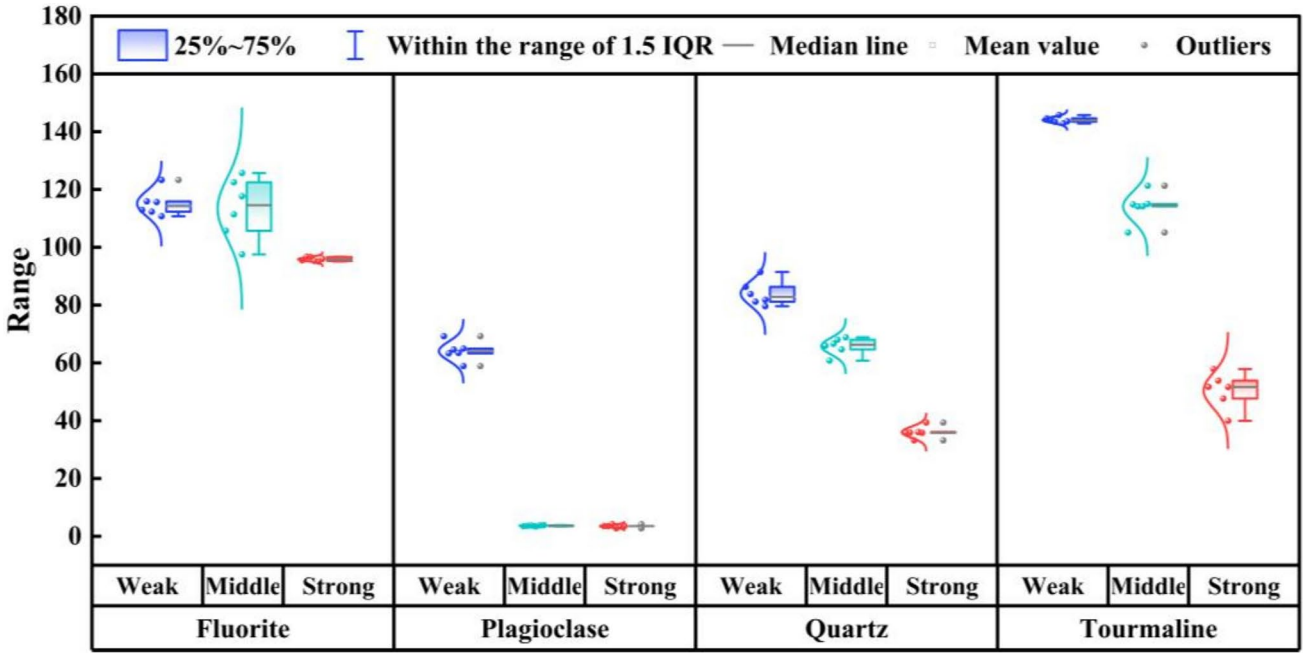


Fig. 20 Statistical distribution of reduced modulus for rock-forming minerals across three laser-induced heat-affected zones. IQR represents the interquartile range

their reduced modulus. This finding aligns with the general consensus that reduced modulus correlates positively with contact hardness, an observation further supported by the experimental results in this study. As illustrated in Table 2 and Fig. 20, minerals T and F, which exhibit higher hardness, also possess higher reduced modulus. Specifically, in the weakly affected zone, the reduced modulus for T and F are 144.07 GPa and 115.15 GPa, respectively. These values transition to 114.12 GPa and 113.43 GPa in the moderately affected zone, and drop to 50.45 GPa and 95.89 GPa in the strongly affected zone. Mineral Q follows with a reduced modulus ranging from 36.02 GPa to 84.05 GPa, while mineral PL exhibits the lowest values, distributed between 3.52 GPa and 64.08 GPa.

Notably, the reduced modulus of mineral F shows only a marginal decline, mirroring its hardness stability. This resistance to thermal degradation may be attributed to the buffering effect of surface impurities or inclusions. Conversely, mineral T exhibits a substantial reduction in reduced modulus, consistent with its hardness variation under laser treatment. This is hypothesized to stem from the surrounding effect of melt generated during laser irradiation.

To quantify the degradation of the reduced modulus relative to increasing laser thermal intensity, the percentage decreases between adjacent zones were calculated. From the WAZ to the MAZ, the reduced modulus decreased by 1.49% (F), 20.79% (T), 21.70% (Q), and 94.29% (PL). When comparing the WAZ to the SAZ, the total reductions further intensified to 16.72% (F), 57.14% (Q), 64.98% (T), and 94.51% (PL). These results indicate that mineral PL undergoes the most significant weakening (exceeding 94%), followed by T and Q, while mineral F maintains the highest thermal stability with a maximum reduction of less than 17%.

Overall, as the intensity of laser-induced thermal influence increases, the key mechanical parameters (hardness and

reduced modulus) of the four constituent minerals exhibit a consistent downward trend. The distinct mechanical responses of these minerals to high-temperature laser treatment underscore the unique structural and compositional characteristics of each mineral phase. In addition, the gradual decrease in nanomechanical properties from the strongly affected zone to the weakly affected zone is consistent with the radial variation in surface characteristics used for zoning.

Notably, Q underwent significant mechanical degradation following laser irradiation, which may be attributed to the thermal weakening of its silicon-oxygen tetrahedral structure. Molecular dynamics simulations conducted by He et al. (2025) revealed that as temperatures rise from 300 to 800 K, the Si–O bond lengths and bond angles in α-quartz undergo substantial deformation. This micro-structural alteration leads to a marked reduction in fundamental mechanical properties, such as the elastic and bulk moduli, with observed decreases ranging from 6.87% to 34.05%. The experimental findings in this study align with these theoretical insights, suggesting that the laser-induced softening of Q is driven by thermal lattice distortion.

To further evaluate the consistency of the nanoindentation results, hardness values obtained from the weakly affected zone, reflecting the near-unirradiated mechanical state of each mineral, were compared with literature data (Table 3). The measured hardness values generally fall within the reported range, although variations exist among studies due to differences in mineral properties, crystallographic orientation, surface preparation, and nanoindentation testing parameters. Note that the modulus-related parameters in previous studies mainly represent elastic modulus, whereas this study reports reduced modulus obtained from nanoindentation measurements. Therefore, these values are not directly equivalent and are only compared to indicate the general magnitude of the mechanical properties of the investigated minerals.

Table 3 Hardness and modulus-related parameters of representative minerals obtained from nanoindentation tests

Mineral	Hardness /GPa	Hardness (this work)[a]/GPa	Elastic modulus/GPa	Reduced modulus (this work)[a]/GPa	References
Fluorite (F)	2.37 ± 0.02	2.39 ± 0.12	139.7 ± 3.8	115.15 ± 4.18	Broz et al.(2006)
Plagioclase (PL)	3.2 ± 0.5	5.92 ± 1.45	67.2 ± 6.4	64.08 ± 2.70	Liu et al.(2023)
	7.7 ± 1.2		67.4 ± 6.2		Bai et al.(2024)
	6.9 ± 1.2[b]		56.6 ± 5.3[b]		Bai et al.(2024)
Quartz (Q)	12.45 ± 3.5	10.29 ± 1.31	128.9 ± 41.7	84.05 ± 3.83	Komadja et al.(2022)
	14.5 ± 0.9		106.3 ± 2.9		Liu et al.(2023)
	14.4 ± 2.0		102.2 ± 8.4		Bai et al.(2024)
	13.0 ± 0.7		104.2 ± 5.9		Zhu et al.(2007)
Tourmaline (T)	15.68[c]	13.00 ± 0.81	208.44[c]	144.07 ± 1.81	Adeoye and Adewoye (2006)
	15.29[d]		279.35[d]		Adeoye and Adewoye (2006)

a. Values from this work correspond to the weakly affected zone

b. Altered plagioclase

c. Basal plane

d. Prismatic plane

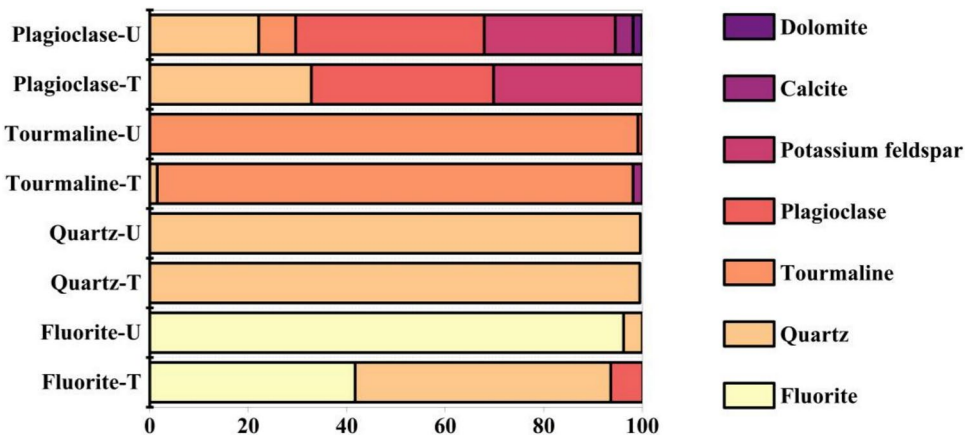


Fig. 21 Evolution of mineral compositions and relative contents of rock-forming minerals after laser irradiation

4 Discussion

4.1 Mineralogical Evolution of Rock-Forming Components Following Laser Irradiation

Figure 21 illustrates the evolution of mineral compositions and relative contents of four mineral samples before and after laser irradiation. The results indicate that the intense heat induced by the laser triggers varying degrees of compositional reconfiguration: the Q phase proportion generally increases, whereas the contents of F, T, and PL exhibit a downward trend. This disparity highlights the distinct thermal stability among different minerals.

In the F samples, the F content plummeted from 96.2% to 41.7% (a reduction of 54.5%), while the Q content surged from 3.8% to 51.9%. This phenomenon aligns with the "laser-induced relative enrichment of the Q phase" observed by Gao et al. (2025). Such a significant phase evolution is likely attributed to the pronounced thermal alteration of F under laser-induced high temperatures. The vigorous smoke emission observed during the experiment suggests that fluorine-bearing components dissipated as volatile fluorides, leading to a substantial decrease in the F proportion. This is further corroborated by the thermogravimetric findings of Ju et al. (2020), which indicated that the volatilization rate of CaF_2 and associated fluorides increases markedly with rising temperature, revealing the inherent susceptibility of F to compositional loss under extreme thermal loads. Furthermore, the trace amount of PL detected (6.5%) may not be a primary mineral phase but rather an artifact of exogenous contamination introduced during sample preparation or experimental procedures.

According to Zheng and Wang (2012), XRD results of Q fibers treated at 600 ~ 900 °C demonstrated that the structure maintained its original amorphous or weakly ordered characteristics without significant phase transformation. The observed reduction in tensile strength was primarily attributed to the proliferation of surface defects rather than crystalline disintegration. Similarly, in the present study, the Q content in the samples increased from 99.6% toward a near-single-phase composition following laser irradiation. Infrared thermography revealed no melt splashing or smoke emission throughout the process. Moreover, nanomechanical testing indicated only a marginal decline in hardness and reduced modulus. These collective results demonstrate that Q exhibits superior thermal stability under intense laser-induced heat, with its microstructure and mechanical load-bearing capacity remaining largely intact.

Under high-temperature laser conditions, the primary T content decreased from 99.1% to 96.6%, accompanied by the disappearance of ancestral PL impurities and the emergence of approximately 1.5% Q. This mineralogical evolution signifies a degree of localized thermal decomposition within the T. The violent smoke emission and melt ejection captured by infrared thermography further substantiate the vigorous thermal response of the samples under laser irradiation. The crystal structure of T hosts volatile constituents, such as boron and fluorine, whose dissipation at elevated temperatures drives the observed mineralogical changes. Additionally, the detected Q may be linked to the localized enrichment of silicon species under high-temperature conditions, leading to the formation of SiO_2-rich phases (Ballirano et al. 2022). While fluctuations in accessory minerals (e.g., feldspar and quartz) might be influenced by sampling representativeness or minor contamination, the overall trend unequivocally reveals the thermal instability and phase reconstruction characteristics of T under laser-induced extreme heat.

In the PL samples, laser thermal action significantly simplified the mineralogy. XRD results showed that K-feldspar content increased from 26.6% to 30.3% and Q from 22.1% to 32.8%, while pre-existing accessory minerals (e.g., calcite and dolomite) essentially vanished. This evolution suggests that high-temperature disturbances promote the decomposition of thermally unstable phases, thereby increasing the relative proportion of stable phases such as Q. Similar mineralogical evolutionary features have been reported by Peng et al. (2024) in their study on the thermal response of Q and feldspar in granite at elevated temperatures.

Further analysis indicates that the reduction in mineral variety in PL samples post-irradiation may be related to its complex initial multi-mineral composition. Under rapid laser heating, the disparity in thermophysical properties between different mineral phases can lead to localized thermal stress concentration, promoting the decomposition or melting of low-stability minerals. Concurrently, the limited sampling scale and inherent heterogeneity of the local mineral composition may also influence the XRD results. In summary, during laser-assisted rock breaking, the mineral composition and inherent thermal stability are critical factors governing localized thermal damage behavior.

4.2 Mechanical Degradation and Weakening Mechanisms of Constituent Minerals across Laser-Induced Thermal Zones

Experimental data (Table 4) indicate that the maximum loads of the four rock-forming minerals undergo varying degrees of attenuation as the laser intensity increases, with the most pronounced degradation occurring in the strongly affected zone. This phenomenon validates the scientific basis

Table 4 Maximum load values of constituent minerals in different heat-affected zones (mN)

Zone	F			PL			Q			T		
	W	M	S	W	M	S	W	M	S	W	M	S
Point-1	210.38	233.1	204.49	307.29	16.06	13.59	445.79	390.52	285.62	500.01	461.92	199.5
Point-2	217.69	159.26	199.66	311.06	15.5	14.52	500.01	370.08	255.08	500.01	500.01	156.36
Point-3	216.82	221.87	205.61	250.19	11.97	7.68	434.62	402.61	250.12	500.01	500.01	215.15
Point-4	248.25	183.23	216.5	340.96	15.5	18.81	456.06	393.4	223.95	500.01	474.41	170.21
Point-5	224.41	180.19	205.01	301.46	15.25	13.66	443.91	336.76	233.91	500.01	493.97	287.03
Point-6	219.39	227.41	198.7	381.35	13.85	14.49	470.86	388.16	250.55	500.01	401.31	131.1

of laser-induced thermal damage zoning from a nanomechanical perspective and indirectly reflects the Gaussian distribution characteristics of laser energy density.

Specifically, F exhibits superior thermal stability, with a load reduction of only 17.18% between the weakly and strongly affected zones. In contrast, Q and T show similar degradation trends, with reductions of 42.88% and 42.60%, respectively. Notably, PL demonstrates extreme thermal sensitivity; its Pmax undergoes a catastrophic collapse when transitioning from the weak to the moderately affected zone, reaching a maximum reduction of 95.07%. Combined with 3D profile scanning analysis, this is attributed to the loose structure of PL, composed of fine-grained aggregates and widespread impurities. These initial structural defects exacerbate interfacial peeling and grain shedding under the thermo-mechanical coupling of the laser.

Statistical analysis (Fig. 22) reveals that the moderately affected zone represents a critical threshold stage for the mechanical weakening of the minerals. This feature is likely associated with thermally activated processes, such as microcrack propagation and the proliferation of crystal defects. As the thermal influence intensifies toward the strongly affected zone the thermal softening effect of the local lattice leads to a further loss of load-bearing capacity. Additionally, the high data dispersion observed in T within the strongly affected zone suggests a complex, non-homogeneous damage evolution under high thermal loads. In summary, the weakening of mineral properties by laser is not a simple monotonic process but a non-linear response driven by the coupling of mineralogical composition and thermal history. This discovery provides a critical basis for evaluating the mechanical behavior of rocks under extreme thermal environments.

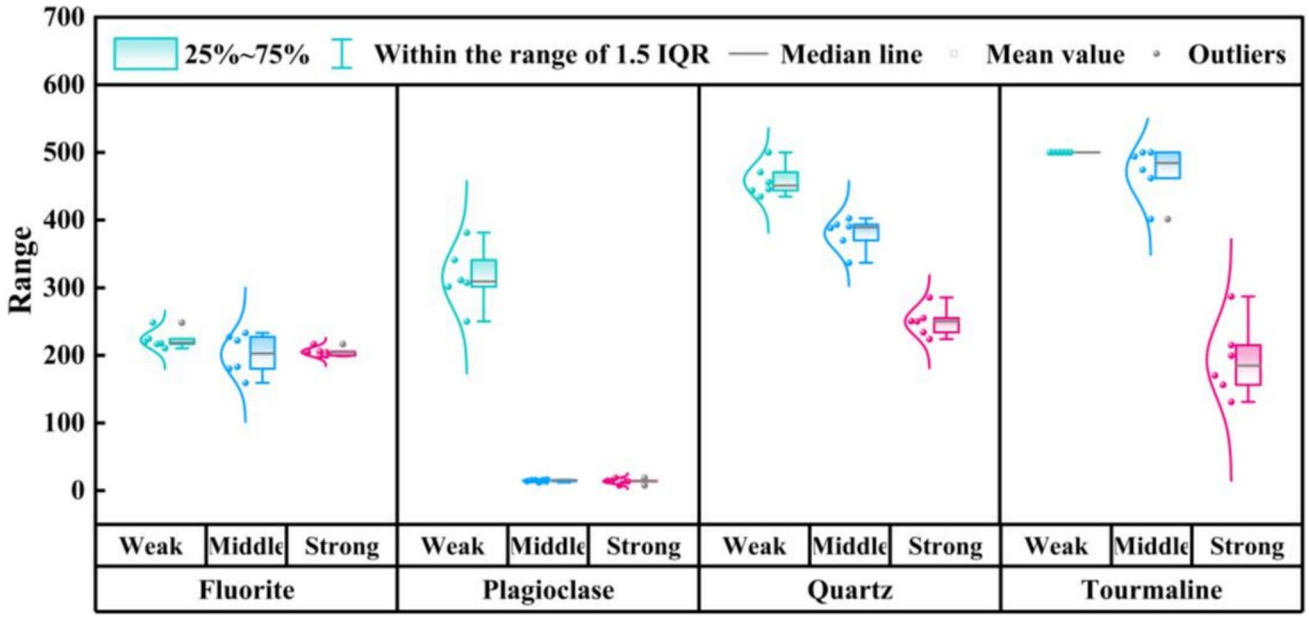


Fig. 22 Statistical distribution of maximum load for rock-forming minerals across three laser-induced heat-affected zones. IQR represents the interquartile range

4.3 Research Limitations and Future Outlook

This study investigated four representative rock-forming minerals found in deep formations: Q, T, PL, and F. Based on the Gaussian distribution characteristics of laser energy, a classification scheme consisting of three heat-affected zones (weak, medium, and strong) was proposed. The degradation laws of these minerals under specific laser parameters were elucidated through multi-scale morphological characterization, compositional evolution analysis, and nano-mechanical testing. The results indicate a high spatial correlation between the surface damage characteristics and the attenuation of micro-mechanical load-bearing capacity. Based on the comparative degradation of hardness and reduced modulus from the weakly affected zone to the strongly affected zone, a preliminary nano-mechanical sensitivity ranking of the four typical rock-forming minerals under laser irradiation was established as: PL > T > Q > F. These results are specific to the single-mineral system investigated under the applied laser parameters (3 kW, 30 s, 15 mm spot size).

This discovery provides a physical foundation for optimizing laser-assisted rock-breaking strategies in deep hard-rock formations. In practical engineering, laser-assisted technology can be prioritized for strata with high concentrations of nano-mechanically sensitive minerals (such as PL and T), utilizing differential thermal stress responses to achieve efficient pre-fracturing, thereby enhancing the efficiency and safety of hard-rock drilling. Despite progress in revealing the laws of laser-induced mineral degradation, several limitations and opportunities for improvement remain:

(1) Refined characterization of micro-zone melt products: Due to the infinitesimal volume and high brittleness of laser-induced melt phases, coupled with the constraints of ablation pit morphology, traditional sample preparation and in-situ mechanical testing struggle to achieve precise data capture. Future research plans to introduce Focused Ion Beam milling and Transmission Electron Microscopy, combined with in-situ nanoindentation, to deeply analyze the microstructural characteristics of the melt zone.

(2) In-depth elucidation of underlying physical mechanisms: Theoretical explanations for the differentiated microscopic responses of minerals require further strengthening. Subsequent work will employ Molecular Dynamics simulations to analyze the molecular-level failure mechanisms of minerals under high-temperature laser action from the perspective of atomic bond energy evolution and lattice dynamics (see Fig. 23), providing a more robust micro-mechanical framework for parameter optimization..

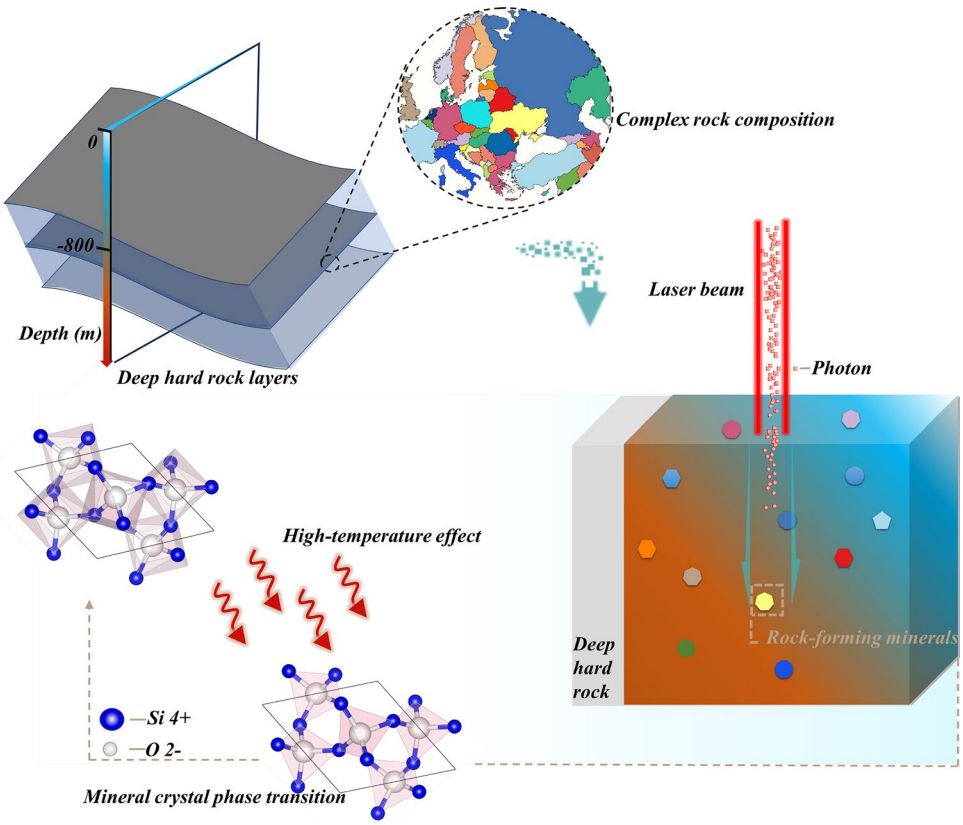


Fig. 23 Schematic of the molecular dynamics and crystallographic interpretation of mineral failure mechanisms induced by laser irradiation

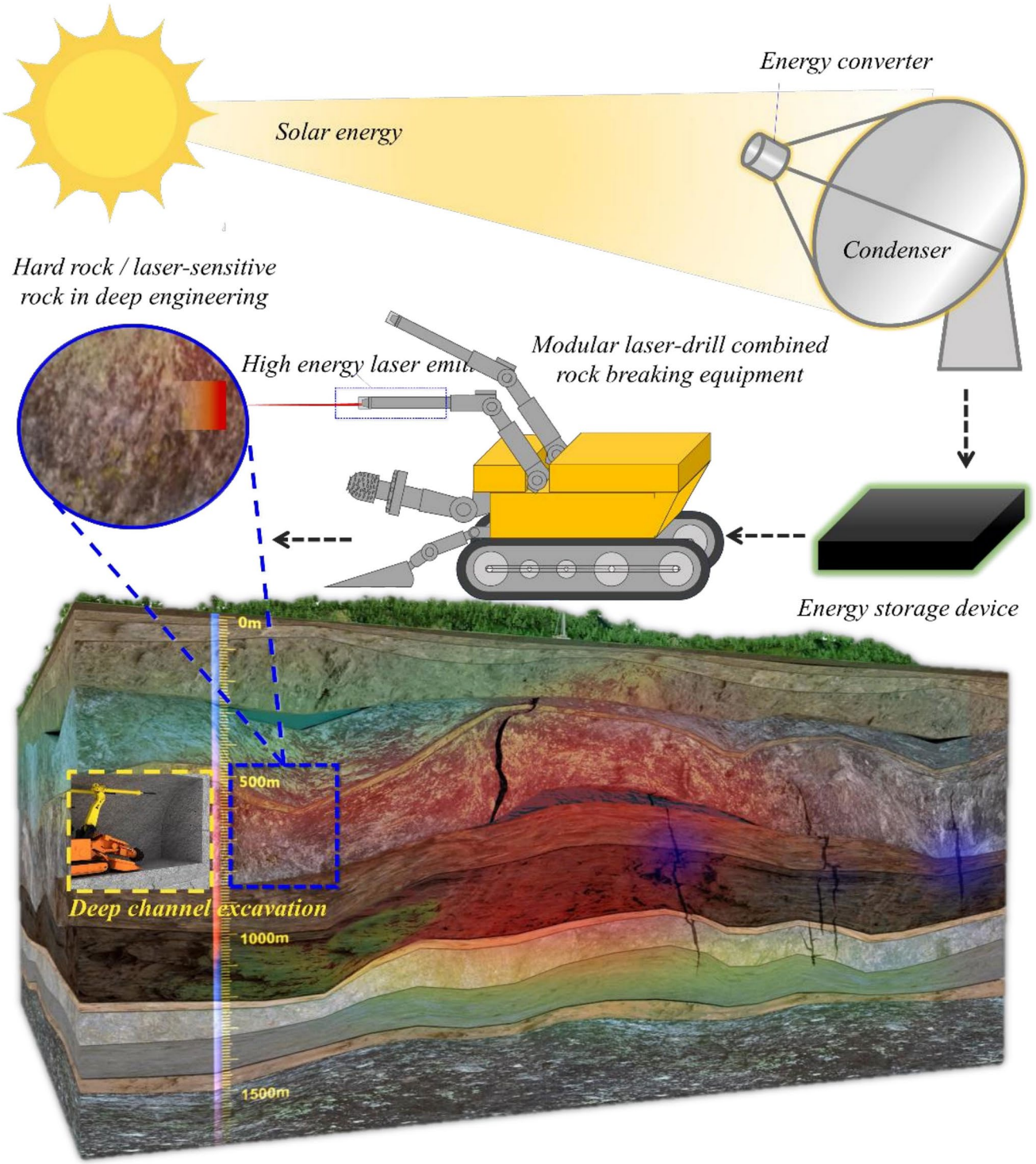


Fig. 24 Conceptual design of a green and efficient vehicle-integrated laser-assisted drilling system for deep formations

In terms of engineering applications and in response to the national "Dual Carbon" strategy, this study proposes a vision for laser-assisted drilling in plateau railway construction. By converting abundant solar energy into laser power, laser-induced pre-fracturing can significantly reduce rock strength and minimize mechanical excavation disturbances, helping to protect the fragile ecosystems of high-altitude regions. Consequently, the author proposes a "green and efficient mobile laser-assisted drilling concept for deep formations" (Fig. 24). This modular, vehicle-integrated solution

enables a stable and self-sufficient supply of clean energy, aiming to provide a safe, green, and efficient drilling solution for remote deep formations where power grid coverage is unavailable.

5 Conclusions

This study investigates the mineralogical heterogeneity of deep hard rocks by systematically analyzing the non-uniform thermal responses and micro-mechanical evolution of F, Q, T, and PL under laser irradiation. By categorizing the laser-affected areas into "weak, medium, and strong" zones, the differentiated mechanisms in temperature response, compositional evolution, and mechanical degradation were elucidated. The following key conclusions are drawn:

(1) Under laser irradiation, F, T, and PL rapidly developed high-temperature focal points (up to 925°C), triggering intense melting and phase-change vaporization. This reveals their low thermal stability and high laser sensitivity. In contrast, Q exhibited superior structural stability, with the thermal field expanding gradually without material ejection. The results confirm that mineral composition is the decisive factor governing the spatiotemporal distribution of the thermal field.

(2) 3D laser scanning indicates that surface damage patterns highly correlate with the Gaussian energy distribution of the laser, showing significant radial attenuation. F and PL underwent severe ablation and amorphization at the spot center, forming glassy microspheres and distinct elevation gradients. T exhibited high-reflectivity melt products and peripheral microcracks reflecting brittle degradation, while Q remained nearly intact, reflecting minimal thermal sensitivity.

(3) XRD analysis revealed a laser-induced "mineral screening effect". While Q maintained high thermodynamic stability, other minerals underwent significant thermal decomposition and selective volatilization. Notably, the primary component of F decreased sharply by 54.5%, and PL experienced compositional simplification with the removal of unstable phases like calcite. An overall increase in the Q phase was observed across all minerals, indicating the enrichment of stable phases under extreme thermal loading.

(4) Nanoindentation tests demonstrate that mechanical properties (hardness, reduced modulus, and maximum load) attenuate as the radial distance from the laser center increases. The mechanical weakening was most pronounced in PL and T, with peak reductions exceeding 80%, whereas Q showed higher mechanical resilience. Furthermore, T exhibited increased load–displacement curve scattering and pop-in events, corroborating its significant surface elevation differentiation and microscopic damage.

Author Contributions Yuanru Li: Writing-original draft, Investigation, Data curation, Methodology. Mingzhong Gao: Conceptualization, Supervision, Funding acquisition, Methodology, Writing—review & editing; Xuemin Zhou: Project administration, Funding acquisition, Writing—review & editing. Yanbo Bai: Investigation, Data curation, Formal analysis. Junjun Liu: Validation, Visualization. Lei Yang: Resources, Data curation. Haichun Hao: Formal analysis, Funding acquisition. Bengao Yang: Investigation, Formal analysis. Jing Xie: Supervision. Zheng Gao: Resources.

Funding This work was supported by grants from the National Science Fund for Distinguished Young Scholars (52225403), the State Key Program of National Natural Science of China (52434004), the National Natural Science Foundation of China (524B2039, 52404365), the Research Team Cultivation Program of ShenZhen University (2023JCT001) and the Shenzhen Science and Technology Program (RCJC20210706091948015).

Data Availability Data have been included in the manuscript.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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